

Detecting departure and destination aerodromes from ADS-B

Version 6.2 — the consolidated study on the datum that ships

EUROCONTROL Performance Review Unit

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Summary

OPDI derives a flight's departure and destination aerodrome from ADS-B state vectors alone. This page consolidates six versions of that work into one account: what the three candidate methods are, how each behaves across its whole parameter space, which settings are best for departures and for arrivals, what the ideas that did *not* work were, and what the pipeline should ship.

The short version is in four parts.

The two roles want different algorithms. Departures are best served by **endpoint** — take the aerodrome nearest the first fix, but only commit when the fix is close and low, and abstain otherwise. Arrivals are best served by **trend**, the algorithm published before 2026. That split holds at every exchange rate tested.

Both roles improve. The recommended configuration — **endpoint** for departures, exact-distance **trend** for arrivals — scores **133,537** against legacy's **122,158** on a three-day sample of 95,116 flights: a gain of **+11,379**, every figure from a real pipeline run.

A better way of choosing among aerodromes is what makes the tuning work. This is the substantive change, and it is worth more than any threshold on this page. The published algorithm narrowed its candidates by **H3 ring count** — an integer stepping about 5.2 km at resolution 7 — and computed exact distance only among the survivors, so an aerodrome a kilometre further out was eliminated before its distance was ever measured. It now carries every candidate through to an exact great-circle comparison and chooses on that.

The difference is not marginal. Applied to arrivals, the same four tuned values are worth **−1,303** under ring selection and **+1,583** under exact distance — the same parameters, the same data, opposite signs. Raising the altitude gate admits samples from higher up, where more aerodromes fall in the same ring, so the coarse rule had most work to do exactly where it was least able to do it. Every tuned arrival value that fails under ring selection gains under exact distance.

trend's altitude cut now measures from the aerodrome, not from sea level. This is what version 6.2 adds. Until it, **trend** gated on a flight level while every other altitude test in the pipeline — **endpoint's**, step 04's ground phase — gated on a height above the aerodrome's own field elevation. The two agree at sea level and diverge by the field elevation everywhere else, so the same nominal ceiling gave a Madrid arrival 2,000 ft less room to vote than an Amsterdam one.

Moving it costs a little accuracy and buys a little coverage, and the aggregate of those two is close to nothing — Section 6.8 explains why it was never going to be otherwise, since most arrivals are at aerodromes below 500 ft where the two datums are identical by construction. The case for the change is not the aggregate: it is that the cut now means the same thing at every aerodrome, and that Section 6.2 shows the gain landing precisely where elevation predicts it should.

1 What this is about

If you have not met this project before, this section is the whole of what you need. Nothing later on assumes another document.

OPDI — the Open Performance Data Initiative — publishes a description of European air traffic derived from what aircraft themselves broadcast. Most airliners transmit their position, altitude and identity several times a second over ADS-B, and a network

of ground receivers records it. That signal is open, but it is raw: a stream of positions with no notion of a *flight*. OPDI's job is to turn it into something you can count with, and to publish the result openly rather than behind a data-sharing agreement.

The flight list is the backbone of that. The pipeline groups each aircraft's positions into **tracks** — one continuous journey, split where the signal stops for long enough — and then gives each track a row: when it started, when it ended, which aircraft, and the two fields this report is about.

ADEP and ADES are the departure and destination aerodromes. They are ICAO codes: EHAM for Amsterdam, LEMD for Madrid. Almost every question anyone asks of flight data needs them — how many movements an airport had, how long a city pair takes, which routes grew. Get them wrong and every count built on top is wrong in a way that is hard to see.

Nothing in the ADS-B signal says where a flight is going. The aircraft broadcasts where it *is*. So the aerodromes have to be inferred from the shape of the trajectory, and that is harder than it sounds:

- Aerodromes cluster. Several of London's lie within a few minutes' flying of one another, and the nearest to a descending aircraft is often not the one it lands at.
- Coverage is uneven. Low-altitude reception is poor in places, so a track may simply stop before it reaches the ground, or start well after take-off.
- Some flights genuinely begin or end outside the observed area, and a method that always answers will confidently name a wrong aerodrome for them.

So this report is about three ways of guessing, and how well each does. Each looks at the trajectory near a candidate aerodrome and decides whether to commit:

- **trend** watches whether the aircraft is climbing or descending near each candidate, and abstains when the altitude trace says neither clearly.
- **endpoint** takes the aerodrome nearest the first or last position, but only commits when that position is close and low; otherwise it abstains.
- **nearest** takes the closest aerodrome unconditionally. It never abstains, and it is included as the external baseline — it is what the widely used **traffic** library does.

Two of the three can refuse to answer, and that is the point. A method that always answers has perfect **coverage** and poor **accuracy**; one that answers only when certain has the reverse. Neither number means much alone, which is why Section 2 sets out how they are traded against one another.

What this version adds. Version 6.2 changes one thing in **trend**: the altitude ceiling below which a position is allowed to vote is now measured **from the aerodrome's own elevation** rather than from sea level. The two are the same at a coastal airport and differ by half a mile at a highland one. Section 6 reports what that was worth.

1.1 A glossary

Terms this report uses without further explanation.

Term	Meaning
ADEP / ADES	The departure and destination aerodrome, as ICAO codes.

Term	Meaning
Track	One continuous run of positions from a single aircraft, split where the signal gaps. The unit each row of the flight list describes.
State vector	A single broadcast position: latitude, longitude, altitude, velocity, at one instant.
H3 cell	A hexagon from Uber’s H3 grid, used to index aerodromes by location so candidates can be found without comparing against every airport in Europe.
Coverage	The share of reference flights the method answered at all. Abstaining lowers it.
Accuracy	Of the flights it answered, the share it got right.
Score, k	Coverage and accuracy pull against each other, so runs are ranked by correct - $k \times$ wrong . k is the exchange rate: how many correct answers one wrong answer is worth. This report uses $k = 2$ throughout and Section 9 shows what happens at other values.
Out-of-area	A flight that genuinely began or ended outside the observed region. Saying so is a <i>correct</i> answer, not an abstention.
Abstention	The method declining to name an aerodrome. Distinct from being wrong, and distinct from out-of-area.
Ground truth	EUROCONTROL Network Manager flight data, used as the reference. Not published with this report; see Section 20.

Reading this page

Every number below is an inline expression over a CSV in `data/`, and every one of those CSVs is produced by a job in `benchmarks/regenerate_v62.py` — which this page runs before drawing anything. Nothing is typed into the prose, and nothing is read that the checked-out code did not produce. Section 20 lists the command and commit behind each figure. Where this page disagrees with versions 1–5, it says so and gives the reason; those versions stay published unchanged.

2 How the metrics work

Five quantities are reported for every setting, for each role separately. Four of them are unavoidable and the fifth is a judgement, so it is worth being precise about all five before any of them are used to argue for a parameter.

Write N for the number of ground-truth flights in the sample, a for the number the method *answers* (names an aerodrome, or explicitly says out-of-area), c for the number it answers **correctly**, and $w = a - c$ for the number it answers **wrongly**.

Coverage is a/N : the share of flights the method is willing to commit on. A method that abstains everywhere has coverage zero and cannot be wrong.

Accuracy is c/a : of the flights it committed on, the share it got right. Accuracy says nothing about how often the method declined to answer, so a method can buy accuracy simply by abstaining more.

Correct (c) and **wrong** (w) are the raw counts. They matter because coverage and accuracy generally move in opposite directions — every extra answer a method gives is one it was previously unsure about, so raising coverage tends to lower accuracy — and neither ratio says whether a particular exchange was worth making. The counts do.

Score is $c - kw$. It exists to make the exchange rate explicit: a change to a parameter is worth taking if and only if it wins more than k correct answers for each wrong one it introduces.

The value of k is where judgement enters. This study uses $k = 2$ by default, on the following argument. A **silence** — an abstention — corrupts one movement count, and it is *visible*: it arrives downstream as an explicit null that any consumer can filter on. A **wrong aerodrome** corrupts *two* movement counts, the one it invents and the one it omits, and it is *invisible*: nothing downstream can distinguish it from a correct answer. Two errors for one, plus undetectability, is the case for $k = 2$.

That is an argument, not a measurement, and rankings do move with k . So every headline choice on this page is also scored at $k = 1$ and $k = 3$, and where the ranking changes, the page says so.

2.1 Out-of-area, and why the headline numbers are decomposed

Roughly one flight in six in the sample genuinely begins or ends outside the observed area. Such a flight has no nameable aerodrome, and the correct answer is the literal label **OOA**. This creates a way to score well for a bad reason: a method that labels liberally out-of-area collects those flights for free.

So each side is also reported three further ways — out-of-area recall and precision, which show whether the free wins are being taken honestly, and in-area coverage and accuracy, which restrict to flights whose truth is a real aerodrome and are the numbers to read when comparing naming rules.

Table 1: The reported quantities.

Quantity	Definition	Reads as
Coverage	a/N	willingness to commit
Accuracy	c/a	reliability when committed
Correct	c	flights named right
Wrong	$w = a - c$	flights named wrong
Score	$c - kw$	net value at exchange rate k
OOA recall	OOA hits / OOA in truth	does it recognise what it cannot name?
OOA precision	OOA hits / OOA predicted	is it over-using the OOA label?
In-area coverage	answered / truth is a real aerodrome	commitment with the free wins removed
In-area accuracy	correct / answered, truth a real aerodrome	reliability with the free wins removed

3 The sample

The 2025 period is 5-7 June 2025, three consecutive days, giving **95,116 ground-truth flights** after alignment with the Network Manager reference.

Ground truth is the EUROCONTROL Network Manager flight table, joined to ADS-B tracks on aircraft address, callsign and date. The join key is `AIRCRAFT_ADDRESS`, which is the `icao24` of the ADS-B message.

4 The three methods

All three name an aerodrome by choosing among candidates, and two of the three choose the same way, so the shared machinery is worth stating once:

Candidate selection. Every aerodrome whose H3 detection zone the aircraft passed through is a candidate. Each candidate's exact great-circle distance to the track's closest approach is computed, and the smallest wins. No candidate is eliminated before its distance is measured.

The scheduled-service penalty. An aerodrome without scheduled commercial service has a fixed distance — 10 NM as shipped — added to its effective distance before that comparison, so a military or general-aviation field must be *clearly* nearest rather than merely nearest. It re-ranks candidates without changing how many flights are answered, so it can move accuracy and cannot move coverage. It applies to **both trend** and **endpoint**.

Where the methods differ is the evidence they read and when they refuse to answer.

4.1 The steps

Every box in the diagrams below is one of these.

S1 — Read the track. One aircraft's state vectors for one flight, already split into a track by step 02 and cleaned by step 02a.

S2 — Smooth the altitude. A centred five-sample rolling mean over barometric altitude, computed over the *whole* track before any altitude cut. Smoothing after the cut would truncate the window exactly where altitude is changing fastest, which is the worst place to lose half of it.

S3 — Pre-filter on pressure altitude. A cheap, deliberately loose cut whose only job is to keep the aerodrome join affordable. Its ceiling is the configured cap *plus the highest field elevation in the reference*, so it cannot discard a sample that the exact test at S6 would have kept.

S4 — Join the aerodrome zones. Each sample is matched to candidate aerodromes through an H3 index at resolution 7, then narrowed to those within the detection radius by exact great-circle distance.

S5 — Attach the field elevation. Each candidate aerodrome's elevation, from OurAirports. A candidate whose elevation is unknown gets zero, which reduces the next step to the sea-level behaviour rather than dropping the aerodrome.

S6 — Cut on altitude. *This is the datum-sensitive step.* Samples above the ceiling are excluded from the vote. **trend** now measures that ceiling **above field elevation**; before this version it was a flight level.

S7 — Vote. Each surviving sample votes on the sign of its smoothed altitude change: rising for a departure, falling for an arrival.

S8 — Decide, or abstain. If one direction leads the other by more than the margin, the track is departing or arriving. Otherwise **trend** says nothing.

E1 — Take the endpoint fix. **endpoint** reads one sample, not many: the track’s first fix for a departure, its last for an arrival.

E2 — Test the gates. The fix must be within the radius, and either flagged on the ground or no higher than the height gate **above field elevation**.

E3 — Rank the survivors. Nearest wins, on exact great-circle distance, plus a penalty for aerodromes without scheduled service so that a military or general-aviation field must be *clearly* nearest rather than merely nearest.

E4 — Or say where it came from. Where the gates fail and the fix sits at the edge of the observed area, **endpoint** reports **out-of-area** rather than guessing. Precedence is aerodrome first, border second: Ponta Delgada lies about 8 NM inside the western edge, and the other order labelled its departures out-of-area with the aircraft still on the runway.

4.2 trend — the algorithm published before 2026

For every state vector below an altitude ceiling and within a radius of an aerodrome, **trend** asks whether smoothed barometric altitude is rising or falling. Each vector casts a vote. If climbing votes exceed descending votes by more than a margin, that aerodrome is the departure; if the reverse, the destination. Among the aerodromes that survive that test, the nearest wins.

That ceiling is a **height above the candidate aerodrome’s own field elevation**, and that is what version 6.2 changed. Every published OPDI flight list before it cut on a flight level instead — a height above the standard pressure surface, and so above a datum with no relation to the aerodrome being tested. Section 6 reports what moving it was worth, and Section 4 above marks the affected step as **S6** in each diagram.

Its abstention is evidential: it stays silent when the altitude trace near an aerodrome does not clearly rise or fall. It has **no out-of-area concept** — it names an aerodrome or says nothing — which Section 21 returns to.

Three of its four constants had never been swept before this study: the altitude ceiling, fixed at FL40; the detection radius, fixed at 30 NM; and the datum that ceiling is measured against, which had never been treated as a choice at all. The scheduled-service penalty described above is new to **trend** in version 6; before it, **trend** ranked on raw distance alone.

4.3 trend, arrivals

4.4 trend, departures

The same machinery reading the same samples; only the direction of the winning vote differs. **trend** has **no out-of-area concept** — it names an aerodrome or says nothing — which is why the shipped configuration does not use it for departures.

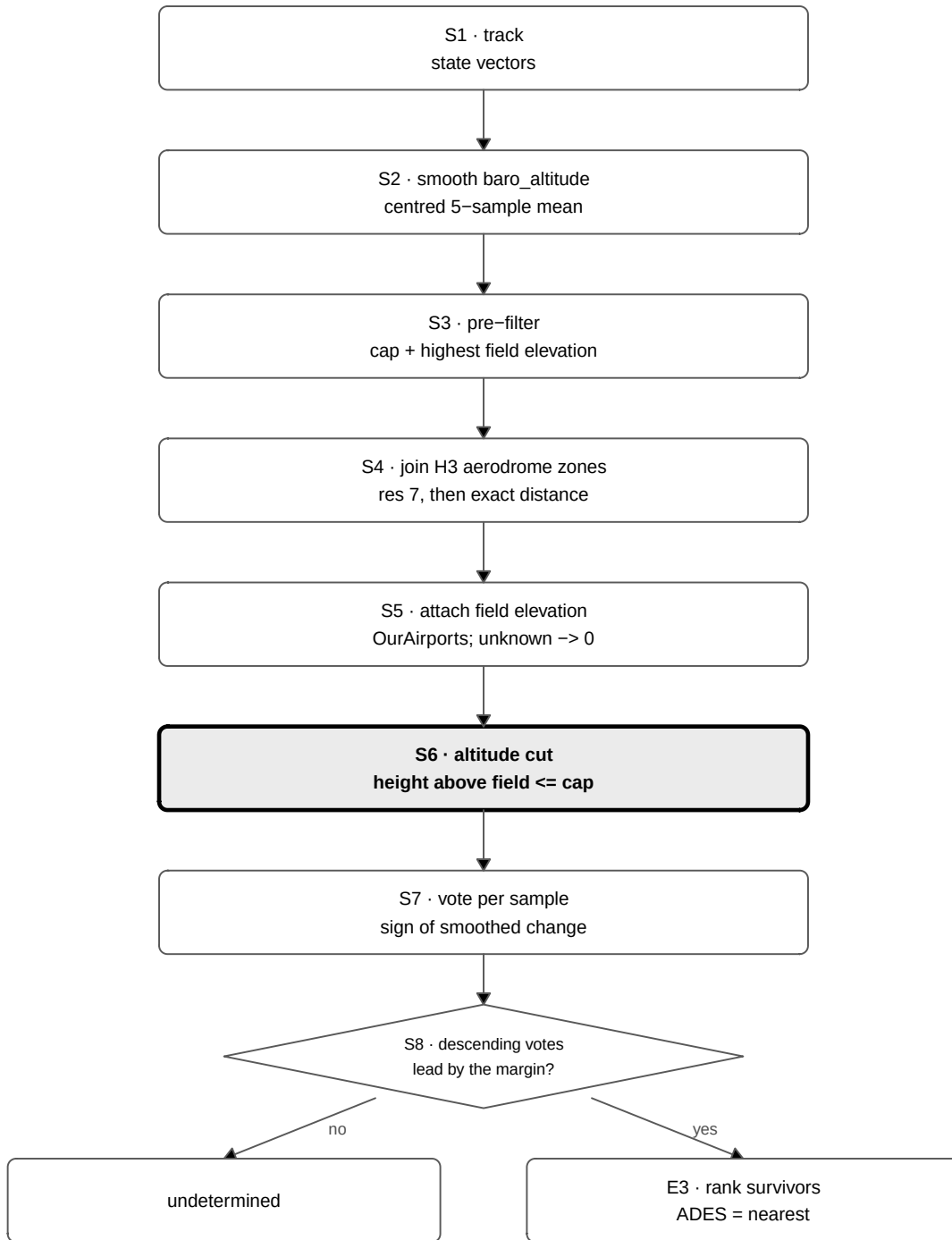


Figure 1: **trend** naming an arrival aerodrome (ADES). S6 is where the two altitude datums differ.

4.5 nearest — the *traffic* library's rule

The reference implementation in the **traffic** library takes the aerodrome nearest the first and last point of the trajectory, unconditionally. It never abstains, so its coverage is total and its accuracy is whatever the geometry gives. It is included as the external baseline.

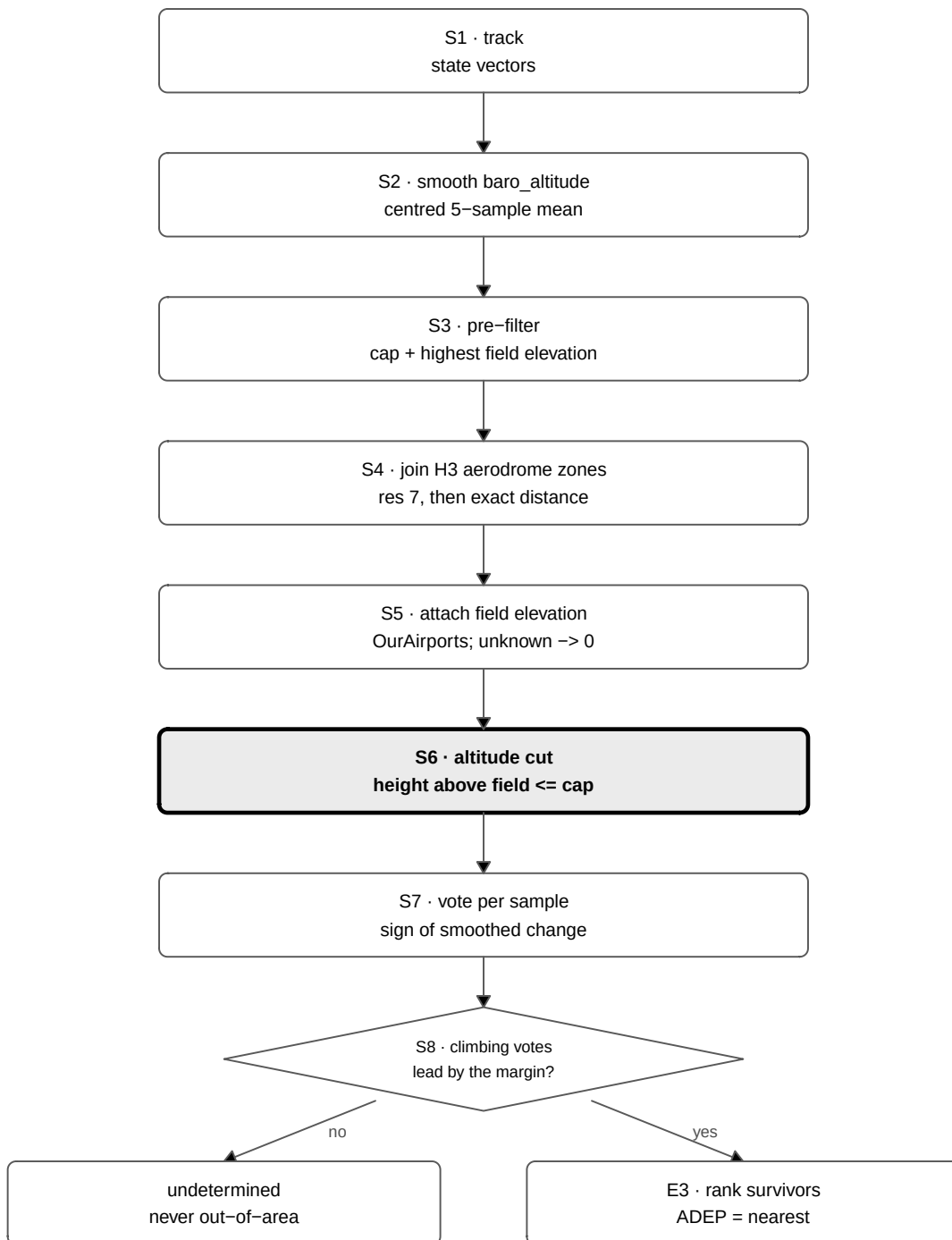


Figure 2: **trend** naming a departure aerodrome (ADEP). The same machinery; only the winning vote direction differs.

4.6 endpoint — nearest, with abstention

endpoint is **nearest** plus two gates: the endpoint must lie within a radius of the aerodrome, and either on the ground or below a height above the field — above *field elevation*, not above sea level, since a fixed altitude cut-off means nothing at an aerodrome sitting at 5,000 ft.

Failing both gates, the method abstains rather than guessing. Its abstention is geometric where **trend**'s is evidential. Unlike **trend** it can also say **out-of-area**: when the failing endpoint sits at the edge of the ingestion region rather than near any aerodrome, the flight came from or went to somewhere outside the observed area, and it says so.

4.7 endpoint, departures

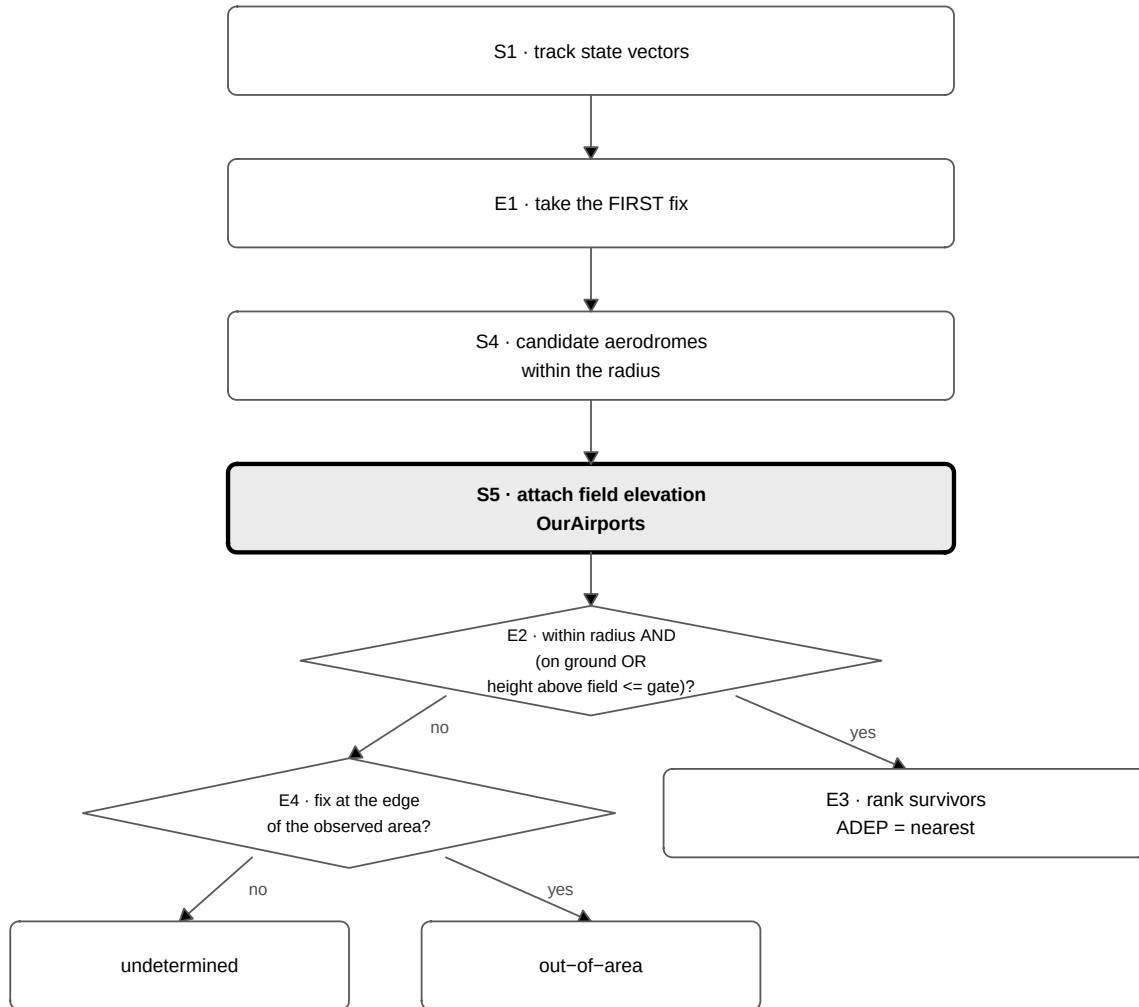


Figure 3: **endpoint** naming a departure aerodrome (ADEP). E2 already measured height above field, and still does.

4.8 endpoint, arrivals

4.9 Where the two datums part company

Only **S6** and **E2** read an altitude, and until this version they read it against different references:

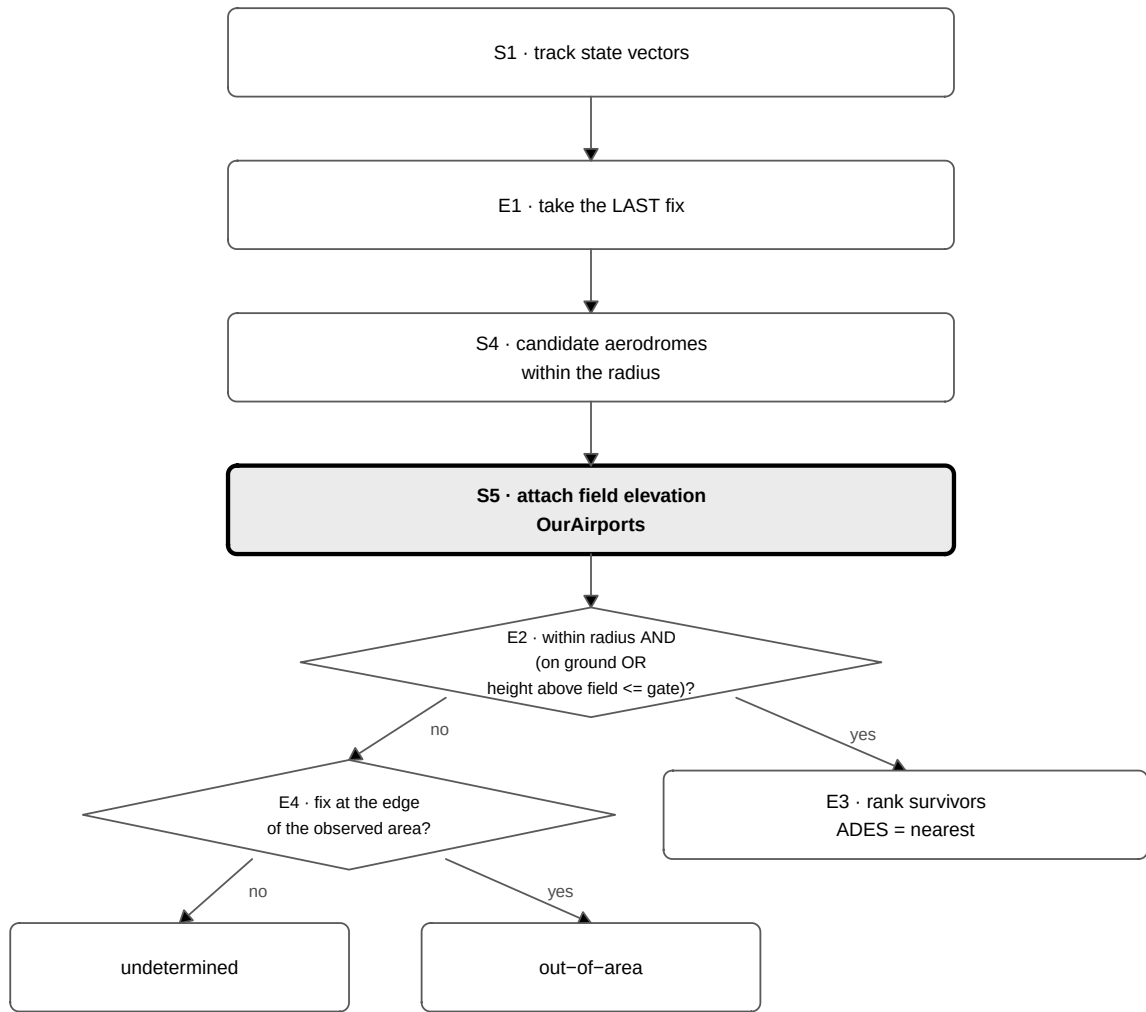


Figure 4: **endpoint** naming an arrival aerodrome (ADES). Identical to the departure flow but reading the track’s last fix.

Test	Threshold	Datum, before v6.2	Datum, from v6.2
trend S6	trend_max_height_ft	flight level (sea level)	above field elevation
endpoint E2	endpoint_height_ft	above field elevation	above field elevation
step 04 ground phase	phase_ground_ceiling_ft	above field elevation	above field elevation
nearest	<i>none</i>	—	—

nearest applies no altitude condition at all, on any datum. That is deliberate: it is the naive **traffic**-faithful baseline the other methods are scored against, and giving it a gate would defeat its purpose.

! A flight level is not a height

FL60 is not the same cut as 6,000 ft. The pipeline derives the flight level with an integer cast, so `flight_level <= 60` admits everything below **6,100 ft**. Every datum comparison in this report therefore holds the ceiling at 6,100 ft above field, not 6,000 — otherwise it would move the ceiling and the datum at the same time and could not attribute the result to either.

The same arithmetic is why the sweep grid in Section 5 carries 6,100 ft rather than 6,000: that is the above-field ceiling admitting exactly the band FL60 admits.

5 Trend: the full parameter sweep

5.1 The untuned baseline

- **ADEP** at the legacy setting (6,100 ft above field, margin 4, 30 NM, no penalty): coverage 72.85%, accuracy 97.90%, 67,835 correct, 1,454 wrong, score 64,927.
- **ADES** at the legacy setting (6,100 ft above field, margin 4, 30 NM, no penalty): coverage 69.70%, accuracy 94.99%, 62,970 correct, 3,324 wrong, score 56,322.

5.2 The best setting found

- **ADEP**: 12,000 ft above field, margin 0, 20 NM, penalty 10 NM — coverage 77.55%, accuracy 98.53%, 72,680 correct, 1,083 wrong, score 70,514 (+5,587 against the untuned baseline).
- **ADES**: 6,100 ft above field, margin 2, 20 NM, penalty 10 NM — coverage 69.76%, accuracy 97.34%, 64,586 correct, 1,765 wrong, score 61,056 (+4,734 against the untuned baseline).

5.3 The full trend grid

`trend` has three geometric parameters, so the sweep is a cube and cannot be printed as one table. It is sliced instead: for each role, score by flight-level cap against radius at that role's best vote margin, then the margin's own profile, then the penalty. Every cell of the cube is in `data/trend_sweep_2025.csv`.

ADEP — score at $k = 2$, by altitude ceiling (rows) and radius in NM (columns), at the winning vote margin of 0. Ceilings are heights above field elevation.

Ceiling	5	10	15	20	30	40	60	80
2,000 ft above field	60,651	60,328	60,134	59,817	59,162	58,612	56,756	54,660
3,000 ft above field	63,690	63,976	63,896	63,783	63,249	62,973	61,875	59,969
4,000 ft above field	65,018	66,554	66,461	66,362	65,779	65,527	64,576	62,878
6,100 ft above field	65,134	68,157	68,203	68,094	67,793	67,405	66,529	65,121
8,000 ft above field	64,921	68,793	69,420	69,336	69,001	68,669	67,769	66,182
10,000 ft above field	64,598	68,779	70,135	70,346	69,893	69,512	68,737	67,322
12,000 ft above field	64,122	68,329	69,988	70,514	70,281	69,917	69,174	68,056

ADES — score at $k = 2$, by altitude ceiling (rows) and radius in NM (columns), at the winning vote margin of 2. Ceilings are heights above field elevation.

Ceiling	5	10	15	20	30	40	60	80
2,000 ft above field	58,677	59,521	59,242	59,063	58,707	58,379	57,476	56,849
3,000 ft above field	58,514	60,341	60,242	60,091	59,824	59,499	58,819	58,316
4,000 ft above field	58,426	60,284	60,743	60,814	60,194	59,800	59,333	58,937
6,100 ft above field	57,641	58,983	59,708	61,056	60,970	60,658	60,024	59,696
8,000 ft above field	56,485	57,809	58,482	59,999	60,167	59,699	58,941	58,573
10,000 ft above field	54,399	55,645	56,303	57,742	58,134	57,811	56,961	56,455
12,000 ft above field	52,742	53,958	54,590	55,744	56,157	55,995	55,389	55,012

The vote margin, at each role’s winning cap and radius:

Role	Margin	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.55%	98.53%	72,680	1,083	70,514
ADEP	2	77.32%	98.59%	72,506	1,039	70,428
ADEP	4	77.12%	98.63%	72,350	1,008	70,334
ADEP	8	76.53%	98.65%	71,810	983	69,844
ADEP	16	75.09%	98.70%	70,488	931	68,626
ADES	0	69.91%	97.24%	64,659	1,832	60,995
ADES	2	69.76%	97.34%	64,586	1,765	61,056
ADES	4	69.36%	97.40%	64,264	1,713	60,838
ADES	8	68.86%	97.48%	63,846	1,649	60,548
ADES	16	67.63%	97.59%	62,773	1,551	59,671

The scheduled-service penalty applied to trend, which the legacy algorithm does not do at all, swept at each role’s winning geometry:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	77.55%	96.26%	71,005	2,758	65,489
ADEP	5	77.55%	98.33%	72,532	1,231	70,070
ADEP	10	77.55%	98.53%	72,680	1,083	70,514
ADEP	15	77.55%	98.39%	72,576	1,187	70,202
ADEP	20	77.55%	98.24%	72,462	1,301	69,860
ADEP	30	77.55%	98.24%	72,462	1,301	69,860
ADES	0	69.76%	95.12%	63,111	3,240	56,631
ADES	5	69.76%	97.04%	64,390	1,961	60,468
ADES	10	69.76%	97.34%	64,586	1,765	61,056
ADES	15	69.76%	97.27%	64,539	1,812	60,915
ADES	20	69.76%	97.17%	64,474	1,877	60,720
ADES	30	69.76%	97.17%	64,474	1,877	60,720

Coverage is constant down each block and only accuracy moves, which is the signature of a pure re-ranking: the penalty changes which aerodrome is named, never whether one is.

6 The altitude datum

trend’s altitude cut is the one threshold in this study that has never been measured against the aerodrome it applies to. Until version 6.2 it read a flight level — a height above the standard

pressure surface — while **endpoint**, the ground phase in step 04, and every other altitude gate in the pipeline read a height above the aerodrome’s own field elevation.

That is a defensible thing to measure and an indefensible thing to leave unexamined, because the two agree exactly at sea level and diverge by the field elevation everywhere else. At Madrid the same nominal ceiling sits 2,000 ft lower above the runway than it does at Amsterdam; at an Anatolian aerodrome, 5,000 ft lower.

This chapter reports what happened when the cut was moved onto the datum every other gate already used. The short answer is that it costs a little accuracy, buys a little coverage, and is adopted for a reason that is not either of those numbers — Section 6.4.

6.1 The two sweeps are one measurement

Everything in this chapter compares an above-field curve against a sea-level curve. That comparison is only worth reading if the two are the same measurement differing in one respect, so this is checked rather than assumed.

They come out of a single vote cache. `trend_sweep_agl.py` counts each track’s votes at every ceiling on **both** datums in one pass, so the two curves are two readings of one table rather than two studies that happen to share a grid. Its parameter grid is identical to the sweep version 6 published — same flight-level caps, same vote margins, same radii, same penalties — with the above-field ceilings added alongside.

The consequence is testable: the sea-level arm should reproduce version 6’s committed sweep exactly. It does.

Across 371 grid cells on each period, **not one value differs**. That is what licensed retiring version 6’s two `trend_sweep` jobs rather than re-running them: they would have recomputed numbers this study already produces. It is also evidence in its own right — extending the harness to a second datum did not perturb the measurement it was extending.

6.2 The band breakdown, read first

This section comes before the headline deliberately. The headline is a single pair of numbers and it is consistent with several explanations; the band breakdown is the only part of this study that can distinguish between them. Read in the other order, it becomes a search for confirmation.

The prediction the design committed to, before any of this was run: **no movement at sea-level aerodromes, and gains concentrated at elevated ones**.

Table 2: Arrivals, 2025 sample. Δ is field minus sea level.

Band	Aerodrome	Truth	Correct MSL	Correct field	Δ	Δ less top mover	Δ less busiest	Δ relative
<500	490	66,612	49,514	49,571	57	33	57	+0.12%
500- 1500	164	15,285	12,593	12,621	28	17	28	+0.22%
1500- 3000	77	4,434	1,919	2,028	109	70	110	+5.68%
>3000	26	705	5	5	0	0	0	+0.00%
unknown	49	8,080	0	0	0	0	0	+0.00%

The effect is where it was predicted to be. Measured against each band’s own base of correct arrivals — which is the honest denominator, since the bands differ in size by two orders of magnitude — the sea-level control band and the band where the change should bite hardest move by very different amounts.

The <500 ft control band moves by 0.12% and the 1,500–3,000 ft band by **5.68%** – a difference of roughly 49-fold between the band where the datum change is arithmetically a no-op and the band where it bites hardest.

It also survives both robustness checks. The census run before this study began showed each treatment band resting on a single aerodrome, so a band mean alone could not separate an elevation effect from a Madrid effect. Removing the largest mover, and removing the busiest aerodrome in the band, both leave it substantially intact.

Removing the largest mover leaves +70 of the +109; removing LEMD, the busiest field in the band, leaves +110 – very slightly *more*. The result is not one aerodrome’s result.

i The control that had to hold, and did

Departures move by exactly zero in every band. The shipped configuration takes departures from **endpoint**, which never reads **trend**’s altitude cut — so a non-zero number anywhere in that row would have meant this change was touching something it has no business touching. It is the cheapest available check that the intervention is the one described.

6.3 What it costs

Table 3: Arrivals through `process_dai` itself, 2025 sample.

Run	Coverage	Accuracy	Correct	Wrong	Score (k=2)
datum_msl	68.29%	98.58%	64,031	923	62,185
datum_field	68.60%	98.42%	64,225	1,028	62,169
legacy	66.99%	98.19%	62,564	1,151	60,262

Moving to the field datum buys **194 more correct arrivals and 105 more wrong ones**. Coverage rises 0.31 pp; accuracy falls 0.15 pp; the score at $k = 2$ moves from 62,185 to 62,169 — 16 points the wrong way. In net terms it is close to a wash, tilted toward answering more flights and being fractionally less certain about each.

6.4 Why it ships anyway

The adoption criteria fixed in the design were: coverage up by at least 0.5 pp, accuracy not falling, and the gain concentrated at elevated fields. **The third holds and the first two do not.** On a cost-benefit reading, that would be the end of it.

The change is adopted regardless, and the reason is worth stating plainly because it is not a benchmark result.

A ceiling measured from sea level is *not the same test at different aerodromes*. At Amsterdam it admits 6,000 ft of climb and descent; at Ankara, 2,975; at Erzurum, roughly 240. Nobody chose that gradient — it is an artefact of which datum the threshold happened to be written against,

and it makes the method’s willingness to answer depend on the terrain under the aerodrome rather than on the evidence in the trajectory. **endpoint** was corrected on exactly this reasoning in version 6, and step 04’s ground detection in the same release. **trend** was the last one left.

So this is a **correctness fix, not an optimisation**, and it is not the benchmark’s place to veto it. What the benchmark is for is telling us what the fix costs, which it has done: a third of a percentage point of coverage gained, a sixth of a point of accuracy given up, and the movement concentrated where the mechanism predicts. That is the honest accounting, and it is reported here rather than buried because a change adopted on principle still has to declare its price.

The measured null in criteria 1 and 2 does carry one design implication: the ceiling itself, inherited from a sweep conducted on the datum now abandoned, is very likely no longer optimal. Section 21 returns to that.

6.5 The band that should have gained most, and did not

The >3000 band is the one where a fixed sea-level ceiling ought to hurt worst, and it shows no change at all.

5 correct arrivals out of 705, on both datums – a delta of +0.

That is not a null result about the datum. It is **trend** failing at those aerodromes for a different reason:

a recall of 0.71% is not a method being mis-tuned, it is a method receiving almost nothing to work with.

The likeliest cause is ADS-B coverage over the Anatolian plateau, where most of this band’s traffic sits; an aerodrome the feed barely sees cannot be named on any datum.

This matters for how the change should be argued elsewhere. The motivating example in the design was Erzurum at 5,763 ft, where a FL60 cut leaves about 240 ft in which to vote. That example is arithmetically correct and doubly unrepresentative: such fields are rare in the sample, **and** they sit in the one band where the method is independently broken. The population that actually benefits is the far less dramatic 1,500–3,000 ft — Madrid, Ankara, Sofia, Tbilisi — where a FL60 cut still leaves three to four thousand feet of usable climb and descent.

6.6 The ceiling survives the move

FL60 was tuned by version 6 *on the datum version 6.2 abandons*, so there was no reason its optimum should have survived the move. The two sweeps, read out of the same vote cache, settle whether it did.

Table 4: Arrivals, shipped geometry (30 NM, margin 0, penalty 10 NM). Both curves swept out of the same vote cache.

Datum	Ceiling	Coverage	Accuracy	Score
above field	2,000 ft	65.73%	97.92%	58,625
above field	3,000 ft	67.31%	97.78%	59,748
above field	4,000 ft	68.64%	97.33%	60,067
above field	6,100 ft	70.35%	96.98%	60,857
above field	8,000 ft	71.45%	96.11%	60,026
above field	10,000 ft	72.84%	94.53%	57,918

Datum	Ceiling	Coverage	Accuracy	Score
above field	12,000 ft	73.84%	93.21%	55,924
flight level	FL20	64.87%	98.12%	58,225
flight level	FL30	66.92%	97.95%	59,750
flight level	FL40	68.14%	97.78%	60,492
flight level	FL60	70.01%	97.18%	60,952
flight level	FL80	71.18%	96.36%	60,309
flight level	FL100	72.53%	94.89%	58,409
flight level	FL120	73.68%	93.47%	56,354
flight level	FL150	74.47%	91.95%	53,721
flight level	FL200	75.45%	89.54%	49,248

Each datum peaks at the equivalent ceiling — 6,100 ft above field, FL60 on sea level — and 6,100 ft wins at *every* radius from 20 NM outward, production’s 30 NM included. So the ceiling survived the change of datum: it is tuned on the new datum rather than inherited from the old one.

i What this does not say

It does not say the shipped ceiling is optimal. `trend_max_height_ft` ships at **6,000 ft**, and 6,000 ft is not on this grid — the above-field caps were chosen to bracket 6,100, the height equivalent of FL60, because that is what the *datum* comparison had to hold fixed to isolate one variable. A sweep that never evaluated the shipped value cannot endorse it. The two are 100 ft apart and the difference is very unlikely to matter, but “very unlikely to matter” is a prediction, not a measurement. Section 12 carries both through `process_dai` and reports which the pipeline prefers.

Two honest limits. The grid is coarse near the peak (4,000 → 6,100 → 8,000), so the true optimum could sit anywhere in that span; and these are harness numbers, which can differ from the pipeline’s in level even when they agree in shape – the harness re-implements the vote rule over a cached table, and Section 12 measures the gap directly rather than assuming it away. What the sweep establishes is the *shape*: an interior optimum, falling away on both sides, at the value already shipping.

6.7 And it replicates on the second period

Table 5: 2024 sample, above-field datum, arrivals at the shipped geometry.

Ceiling	Coverage	Accuracy	Score
2,000 ft	57.62%	96.74%	48,234
3,000 ft	59.80%	96.68%	49,975
4,000 ft	62.07%	96.70%	51,894
6,100 ft	64.19%	96.19%	52,765
8,000 ft	65.52%	95.41%	52,439
10,000 ft	67.09%	93.53%	50,178
12,000 ft	68.15%	92.06%	48,174

6,100 ft is the optimum on 2024 as well, at every radius from 20 NM outward — the same

value, the same interior peak, on an independent sample a year apart. A ceiling that survives that is tuned, not fitted.

The datum comparison replicates too, including its sign.

The best above-field cell scores 0.02% below the best sea-level cell on 2025 and 0.20% below on 2024. Both are fractions of a percent, and both are diluted for the reason Section 6.8 gives. What matters is that the second period tells the same story rather than a different one — the finding is the algorithm’s and not the sample’s.

i

Why the second period is sweeps only

The 2024 tracks pre-date this study’s H3 indexing, and `process_dai` reads `h3_res_7` off the track table directly, so the pipeline arms cannot run here at all. Version 6 met the same wall and ran the pipeline on 2025 alone. The sweeps get around it because their vote cache is built with the index computed at read time.

So the second period confirms the *ceiling* and the *datum comparison*, which is what it is able to confirm. It does not independently reproduce the band breakdown, and this report does not claim that it does.

6.8 Why the aggregate was never going to show much

The two curves are also nearly identical in level — 58,625 against 58,225 at the shipped geometry, and 61,056 against 61,069 at each datum’s own best cell. That looks like a change doing nothing, and it is worth being precise about why it is not.

70.03% of the ground-truth arrivals sit in the <500 ft band, where the two datums are the same test by construction.

A change that cannot move the majority of the sample cannot move an average over it, however well it works on the rest. The aggregate score is diluted by design, and judging a targeted correctness fix by it is using the wrong instrument — the reading is dominated by the aerodromes the fix was never about.

That is precisely why Section 6.2 exists and why it is read first. The signal is in the 1,500–3,000 ft band, and the aggregate is where that signal goes to be averaged away.

7 Endpoint: the full parameter sweep

endpoint has two geometric parameters. The **radius** is how close to an aerodrome the trajectory’s first or last point must be; the **height** is how far above field elevation it may be. Failing either, the method abstains.

The grid below is the whole sweep, scored for each role. The height row labelled *unrestricted* removes the height gate entirely, which collapses **endpoint** into **nearest** within the radius — so the last row of each table is the cost of having no abstention at all.

ADEP — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	28,283	28,211	28,187	28,163	28,160	28,154	28,154	28,154	28,154
500	39,134	39,040	38,984	38,946	38,943	38,935	38,935	38,935	38,935

height (ft)	5	10	20	30	40	60	80	100	110
1,000	52,179	52,069	51,989	51,943	51,941	51,923	51,923	51,923	51,923
2,000	61,667	61,532	61,438	61,373	61,365	61,321	61,317	61,317	61,317
5,000	65,575	67,910	67,766	67,661	67,588	67,480	67,453	67,453	67,453
8,000	65,704	69,388	69,908	69,726	69,621	69,509	69,474	69,472	69,472
10,000	65,683	69,583	70,964	70,785	70,635	70,495	70,448	70,446	70,444
15,000	65,663	69,601	71,754	71,881	71,674	71,402	71,349	71,347	71,345
20,000	65,628	69,466	71,431	71,523	71,307	70,916	70,833	70,819	70,817
25,000	65,582	69,301	70,956	70,638	69,793	68,975	68,823	68,800	68,796
30,000	65,546	69,167	70,331	69,349	67,962	66,623	66,421	66,315	66,310
35,000	65,449	68,511	68,546	66,102	63,376	60,747	60,381	60,146	60,101
40,000	65,375	68,155	67,349	64,085	60,210	56,179	55,424	54,902	54,663
unrestricted	65,371	68,139	67,301	64,013	60,101	56,037	55,268	54,726	54,477

ADES — score at $k = 2$, by radius (NM, columns) and height (rows)

height (ft)	5	10	20	30	40	60	80	100	110
0	31,630	31,550	31,495	31,447	31,428	31,409	31,405	31,403	31,403
500	49,660	49,569	49,462	49,412	49,391	49,366	49,362	49,356	49,356
1,000	54,156	54,046	53,929	53,877	53,856	53,821	53,813	53,807	53,807
2,000	55,316	56,314	56,186	56,112	56,069	56,026	56,000	55,994	55,994
5,000	55,188	57,285	59,041	58,989	58,898	58,809	58,736	58,728	58,728
8,000	55,207	57,258	59,490	59,776	59,612	59,518	59,437	59,429	59,429
10,000	55,154	57,160	59,432	59,987	59,880	59,792	59,709	59,701	59,701
15,000	55,049	56,935	58,854	59,151	59,029	58,920	58,835	58,827	58,825
20,000	54,955	56,682	57,979	57,288	56,892	56,725	56,622	56,592	56,590
25,000	54,934	56,579	57,372	56,142	55,471	54,931	54,744	54,699	54,697
30,000	54,888	56,447	56,773	55,023	53,895	52,938	52,493	52,424	52,416
35,000	54,403	55,231	53,876	50,303	47,695	44,632	43,182	42,549	42,349
40,000	54,220	54,567	51,927	46,886	43,033	37,446	35,267	33,544	32,986
unrestricted	54,218	54,551	51,882	46,779	42,792	36,997	34,782	33,009	32,437

7.1 Where each role’s optimum sits

- **ADEP**: 30 NM, 15,000 ft — coverage 80.03%, accuracy 98.14%, 74,709 correct, 1,414 wrong, score 71,881.
- **ADES**: 30 NM, 10,000 ft — coverage 68.44%, accuracy 97.38%, 63,397 correct, 1,705 wrong, score 59,987.

Both optima are interior to the grid in both parameters, so neither is an artefact of where the sweep stopped.

8 The scheduled-service penalty

Both methods choose among surviving candidates by distance. Near a major aerodrome that often means choosing between the airport itself and a small airfield a few miles away that almost never sees commercial traffic. The penalty adds a fixed distance to every candidate

without scheduled service, biasing the tie-break toward the aerodrome more likely to be the real one.

It is a pure re-ranking: it changes *which* aerodrome is named, never *whether* one is named. So coverage is flat across the whole sweep and only accuracy moves — which is exactly what makes it cheap.

Endpoint, at 40 NM and 15,000 ft:

Role	Penalty (NM)	Coverage	Accuracy	Correct	Wrong	Score
ADEP	0	80.27%	97.31%	74,299	2,052	70,195
ADEP	5	80.27%	97.71%	74,597	1,751	71,095
ADEP	10	80.27%	97.96%	74,790	1,558	71,674
ADEP	15	80.27%	97.88%	74,724	1,621	71,482
ADEP	20	80.28%	97.65%	74,559	1,796	70,967
ADEP	30	80.27%	97.55%	74,476	1,870	70,736
ADES	0	70.98%	94.61%	63,881	3,637	56,607
ADES	5	70.98%	95.05%	64,171	3,344	57,483
ADES	10	70.96%	95.82%	64,673	2,822	59,029
ADES	15	70.92%	95.59%	64,487	2,973	58,541
ADES	20	70.91%	95.50%	64,414	3,035	58,344
ADES	30	70.87%	95.48%	64,361	3,050	58,261

9 How much the exchange rate matters

k is a judgement, so the honest thing is to show what it decides. The table below re-runs the argmax at $k = 1, 2$ and 3 for both methods and both roles.

k	Role	Best trend	trend score	Best endpoint	endpoint score	Winner
1	ADEP	12,000 ft above field, margin 0, 20 NM, pen 10	71,597	40 NM, 20,000 ft, pen 10	73,363	endpoint
1	ADES	6,100 ft above field, margin 2, 30 NM, pen 10	62,889	40 NM, 15,000 ft, pen 10	61,851	trend
2	ADEP	12,000 ft above field, margin 0, 20 NM, pen 10	70,514	30 NM, 15,000 ft, pen 10	71,881	endpoint
2	ADES	6,100 ft above field, margin 2, 20 NM, pen 10	61,056	30 NM, 10,000 ft, pen 10	59,987	trend
3	ADEP	10,000 ft above field, margin 0, 20 NM, pen 10	69,520	20 NM, 15,000 ft, pen 10	70,568	endpoint
3	ADES	4,000 ft above field, margin 2, 15 NM, pen 10	59,522	30 NM, 8,000 ft, pen 10	58,431	trend

The **choice of method is the same at every k tried**: departures want **endpoint** and arrivals want **trend** whether a wrong answer is priced at one silence, two or three. That is the most robust finding on this page, and it does not depend on the judgement that fixes k .

The parameter *values* are less stable than the method choice, and in a direction that makes sense: a higher k prices wrong answers more dearly, so the argmax moves toward tighter gates

that answer less often and are right more often when they do. Where a headline value moves with k , the justification table records it.

10 Negative results, re-measured

Ideas that were tried and did not work. All are re-run here against current code rather than quoted from the version that first reported them.

10.1 The approach cone

The box rule treats radius and height as independent, and they are not: an aircraft descending toward an aerodrome trades height for distance at roughly a constant rate. A 3° glideslope is 318 ft/NM and the “three times the flight level” rule of thumb about 333 ft/NM, so the endpoints that genuinely belong to an aerodrome should lie in a *cone*, not a box.

The two shapes disagree in both corners. Close and high is inside the box and outside the cone — an aircraft at 15,000 ft overhead is overflying, not arriving. Far and low is outside the box and inside the cone — an arrival whose reception dies at 8,000 ft and 30 NM is on profile.

Replacing the box with a cone is worse for both roles, at every slope swept.

- **ADEP**: best cone (1000 ft/NM, 40 NM) scores 64,988 against the box’s 71,881, $-6,893$. It is worse on both counts — fewer correct *and* more wrong, so no exchange rate rescues it.
- **ADES**: best cone (500 ft/NM, 40 NM) scores 60,095 against the box’s 59,987, $+108$. It is more restrictive, giving up 178 correct to avoid 143 wrong — a trade worth taking only if a wrong answer cost more than 1 silences.

The cone loses for both roles at every slope swept, and the reason is the one that sank every other rate-based idea in this study: the OpenSky archive carries positions forward between updates, so a *position* is trustworthy whenever it appears, while anything derived by dividing by elapsed time is not. A glideslope is a rate in disguise.

10.2 The bearing test

Version 5 proposed asking whether the trajectory *points at* the aerodrome: take the track’s course near its endpoint, take the bearing from that endpoint to the candidate, and require the two to agree within some angle. It was the first signal found that could add answers the abstention refused, and it was proposed in four forms — **rescue** (answer anyway when aligned), **veto** (require alignment even when the gate accepts), **replace** (alignment instead of the gate) and **rerank** (choose *which* aerodrome by alignment).

All four are re-measured here against the current **endpoint** baseline.

Variant	ADEP cover- age	ADEP accu- racy	ADEP score	Δ	ADES cover- age	ADES accu- racy	ADES score	Δ
base	80.03%	98.14%	71,881	—	70.15%	96.22%	59,151	—
rescue: gate OR align ≤ 0.1	80.15%	98.02%	71,713	-168	70.30%	96.15%	59,140	-11

Variant	ADEP cover- age	ADEP accu- racy	ADEP score	Δ	ADES cover- age	ADES accu- racy	ADES score	Δ
rescue: gate OR align $\leq$ 0.25	80.23%	97.97%	71,665	-216	70.49%	95.99%	58,985	-166
rescue: gate OR align $\leq$ 1.0	80.65%	97.72%	71,459	-422	71.05%	95.70%	58,870	-281
rescue: gate OR align $\leq$ 3.0	81.84%	96.87%	70,534	-1,347	72.53%	94.63%	57,868	-1,283
veto: gate AND align $\leq$ 0.1	2.64%	93.32%	2,010	-69,871	2.75%	89.45%	1,788	-57,363
veto: gate AND align $\leq$ 0.25	5.05%	96.15%	4,245	-67,636	5.00%	93.73%	3,861	-55,290
veto: gate AND align $\leq$ 1.0	17.36%	98.47%	15,753	-56,128	15.74%	97.52%	13,856	-45,295
veto: gate AND align $\leq$ 3.0	39.96%	99.04%	36,920	-34,961	37.66%	98.52%	34,232	-24,919
replace: align $\leq$ 0.1 only	2.77%	90.00%	1,842	-70,039	2.90%	88.14%	1,777	-57,374
replace: align $\leq$ 0.25 only	5.25%	93.57%	4,029	-67,852	5.34%	90.91%	3,695	-55,456
replace: align $\leq$ 1.0 only	17.98%	96.56%	15,331	-56,550	16.65%	95.24%	13,575	-45,576
replace: align $\leq$ 3.0 only	41.77%	96.51%	35,573	-36,308	40.05%	95.50%	32,949	-26,202
rerank by alignment, then gate	51.17%	86.58%	29,080	-42,801	46.12%	88.87%	29,214	-29,937
rerank, gate OR align $\leq$ 0.1	52.43%	84.63%	26,875	-45,006	47.44%	86.94%	27,441	-31,710
rerank, gate OR align $\leq$ 0.25	54.09%	82.25%	24,052	-47,829	49.02%	84.47%	24,900	-34,251

The best variant of any kind is **base**, worth +0 on arrivals — and +0 on departures, which is the role **endpoint** is actually used for. Nothing here is worth adopting.

The three variants that let bearing *remove* or *choose* answers are not close: the best of them gives up -34,961 on departures. Alignment is far too blunt to name an aerodrome — at 40 NM a runway subtends a couple of degrees, but so does every other aerodrome behind it on the same radial, and a track on final approach points at all of them equally.

This supersedes version 5's recommendation. That version scored the bearing rescue against **endpoint** alone and never against **trend**, which was already better than both — the comparison that mattered was the one not made.

10.3 Bearing applied to trend

The bearing variants above all modify **endpoint**. **trend** is the algorithm that ships for arrivals, and it fails differently — it has no gate, so it almost always answers, and its errors are misnamings rather than silences. The variant that suits it is therefore a *rerank* or a *tie-break*, not a rescue,

and its candidate set is already narrowed by the vote rule, so alignment is asked a much easier question than on the endpoint side.

Variant	ADES coverage	ADES accuracy	ADES score	Δ vs shipped
base: trend as shipped	68.89%	97.63%	60,871	—
rerank by alignment	68.89%	64.71%	-3,839	-64,710
tie-break by alignment within 0.5 NM	68.89%	97.86%	61,330	+459
tie-break by alignment within 1 NM	68.89%	98.02%	61,642	+771
tie-break by alignment within 2 NM	68.89%	98.06%	61,711	+840
tie-break by alignment within 5 NM	68.89%	96.08%	57,832	-3,039
veto: abstain when align > 0.1 deg	0.18%	100.00%	172	-60,699
veto: abstain when align > 0.25 deg	0.50%	99.58%	473	-60,398
veto: abstain when align > 1 deg	1.24%	99.66%	1,166	-59,705
veto: abstain when align > 3 deg	2.90%	99.38%	2,703	-58,168
veto: abstain when align > 10 deg	8.19%	99.00%	7,556	-53,315

tie-break by alignment within 2 NM improves on the shipped configuration by +840. That is a real gain and it is not currently adopted – it arrived after the defaults were set, and adopting it would mean shipping a value this study has measured once, on one period. It is the first thing a follow-up should confirm.

10.4 The vertical measures

Whether a track is climbing or descending can be read from the broadcast vertical rate, from a two-point altitude slope, or by least squares over the whole window. The three disagree, and version 5 could not say which was wrong.

Window (min)	Freshness	Flights	Broadcast rate vs OLS	2-point slope vs OLS
5	1. <30% fresh	11,905	55.38%	30.34%
5	2. 30-60%	15,499	64.31%	32.82%
5	3. 60-90%	1,104	62.23%	61.78%
5	4. >=90% fresh	720	67.08%	54.31%
10	1. <30% fresh	9,279	55.57%	39.34%
10	2. 30-60%	13,913	63.18%	48.70%
10	3. 60-90%	5,544	75.31%	71.97%
10	4. >=90% fresh	813	64.94%	54.24%
20	1. <30% fresh	7,700	57.25%	39.87%
20	2. 30-60%	10,392	61.80%	48.61%
20	3. 60-90%	11,059	76.26%	71.26%
20	4. >=90% fresh	969	70.59%	60.99%

Across the sample the broadcast rate disagrees with the least-squares slope on 60.66% of cases and the two-point slope on 33.43%; 50.14% of broadcast rates are exactly zero, and 34.63% of samples are fresh. The disagreement is concentrated where position updates are stale, which is the same explanation that sank the approach cone: a position survives the archive’s carry-forward, a rate does not.

11 How the sample is thinned

This is not a negative result — it is a property of the data every other number on this page is computed from, and it is a change to the pipeline that will matter most somewhere else.

OPDI thins the 1 Hz feed to roughly 5 s. The rule this study inherited is fixed-phase — `event_time % 5 == 0`, which keeps a row only if one exists at the single second per window congruent to zero. The rule it now uses keeps the newest row in each 5 s bin, and cannot drop a sample where the modulo rule would.

The bin-based rule is adopted, and **not because it measures better**. It is the sampler the thinning was always meant to be; the modulo rule survives only because the OpenSky archive is a complete 1 Hz grid with carried-forward positions, so a row almost always exists at the congruent second. That is a property of the archive, not a guarantee. If it ever changes, the modulo rule degrades badly and this one degrades gracefully.

For ADEP/ADES detection it should make almost no difference, and it does not: both methods read an altitude *trend* over tens of samples, and one extra fix per bin cannot change the sign of a trend. Where it should matter is the milestones defined by a single instant on the ground — off-block, take-off, landing, in-block — where a recovered bin is the difference between having a fix and not. That measurement belongs to the events study and has not been made here.

What it costs is the question this page can answer, and the answer is: nothing measurable. Each row below compares two complete runs across every **cell** — one cell being one parameter combination — so a difference that appears at a single setting and nowhere else cannot be mistaken for a sampler effect.

Comparison	Role	Cells	Median Δ score	Median Δ coverage	Improved	Worsened
trend sweep	ADEP	371	+6	+0.0021 pp	231	134
trend sweep	ADES	371	+17	+0.0042 pp	310	57
endpoint grid	ADEP	126	-6	+0.0011 pp	41	84
endpoint grid	ADES	126	-12	-0.0137 pp	28	92
pipeline modes	ADEP	6	-10	+0.0000 pp	2	4
pipeline modes	ADES	6	+16	+0.0026 pp	3	3

Across every parameter cell of two complete runs, the largest median coverage difference in either direction is **0.0137 pp**. The trend side is consistently positive — 310 of 371 arrival cells improve, which is a real if tiny effect — and the endpoint side is consistently slightly negative. The shipped configuration takes departures from **endpoint** and arrivals from **trend**, so the two roughly cancel.

Column gloss: a **cell** is one parameter combination; Δ is the bin-based sampler minus the fixed-phase one; **pp** is percentage points of the flights in the reference.

So the sampler is **neutral on accuracy and coverage** for this problem, and is adopted on correctness. It should not be described as an improvement to aerodrome detection.

One consequence has to be stated because it is not reversible by configuration: **track_id values differ** between the two samplers. The rescued rows sit at track boundaries and the splitter breaks on gaps, so a track that gains a fix can be split differently. Re-processing a past month with the bin-based rule will not reproduce the identifiers that were published under the fixed-phase one.

12 What the pipeline actually produces

Everything measured so far has been measured in two stages, and the distinction governs how much any number on this page is worth.

Stage 1 — the research sweep. Hundreds of parameter combinations applied to a cached intermediate table: vote counts per track-aerodrome pair for **trend**, first-and-last-fix candidates for **endpoint**. Cheap enough to be exhaustive, which is what makes a grid of this size possible at all. But it re-implements the decision rule rather than running it, so on its own it can only *propose* a value.

Stage 2 — the production run. The proposed values run through `FlightListProcessor.process_dai` — the same code path that writes the published flight list — and scored identically. No value in Section 18 became a default without passing this stage.

Two differences between the two stages remain, and both are named here rather than left to be discovered in the residual:

1. the pipeline filters zones by radius **band** — the H3 ring the zone was generated at — while the sweep cuts on exact distance;
2. the sweep smooths barometric altitude across the flight-level boundary and the pipeline applies the cut first, so its smoothing window is truncated there.

A third difference, and by far the largest, has been **closed**: the pipeline used to keep only the candidates at the minimum H3 ring count and measure exact distance only among the survivors, so an aerodrome one ring further out was eliminated before its distance was ever computed. It now carries every candidate in the joined disk through to an exact-distance ranking, which is what the sweep always did. That change is the subject of the rest of this section.

Run	Role	Mode	Coverage	Accuracy	Correct	Wrong	Score
legacy	ADEP	trend	67.46%	98.82%	63,406	755	61,896
legacy	ADES	trend	66.99%	98.19%	62,564	1,151	60,262
trend	ADEP	trend	78.04%	97.92%	72,686	1,542	69,602
trend	ADES	trend	71.84%	95.67%	65,373	2,959	59,455
endpoint	ADEP	endpoint	80.03%	98.14%	74,709	1,414	71,881
endpoint	ADES	endpoint	70.15%	96.22%	64,197	2,523	59,151
nearest	ADEP	nearest	91.64%	87.68%	76,422	10,738	54,946
nearest	ADES	nearest	88.55%	79.29%	66,783	17,446	31,891
combined	ADEP	endpoint	79.98%	98.21%	74,707	1,363	71,981
combined	ADES	trend	67.68%	98.54%	63,436	940	61,556
recommended	ADEP	endpoint	79.97%	98.21%	74,707	1,361	71,985
recommended	ADES	trend	68.54%	98.44%	64,177	1,015	62,147

Harness against pipeline, at the same parameters:

Role	Method	Harness score	Pipeline score	Residual	Relative
ADEP	endpoint	71,881	71,881	+0	+0.00%
ADES	trend	61,056	61,556	+500	+0.82%

The two rows could hardly be more different, and the difference is the single most important result on this page.

endpoint reproduces exactly. That is not luck: the endpoint sweep filters the same cached candidate table the pipeline reads, applying the same rule, so harness and pipeline are the same computation and agree to the flight.

trend does not, and the reason is the ranking rule. Both stages now rank candidate aerodromes on exact great-circle distance: the full set of aerodromes whose detection zone the aircraft passed through is carried forward, each candidate’s own closest approach is computed, and the scheduled-service penalty is applied to that distance. Nothing is eliminated before it is measured.

Until this study that was true of the sweep only. The pipeline kept just the candidates at the minimum H3 *ring count* — an integer stepping about 5.2 km at resolution 7 — and applied the tie-break, and the penalty, within that surviving set alone. An aerodrome one ring further out was gone before the penalty could promote it, so a penalty tuned against exact distance did something weaker in production. The residual above is what remains once that rule is the same on both sides: the radius band and the truncated smoothing window, and nothing else.

12.1 The trend tuning re-walked inside the pipeline

The path below runs through `process_dai` itself. Each row **adds one parameter change to the row above it** and rebuilds the whole flight list with everything adopted so far, so row n differs from row $n-1$ in exactly one value. It answers which of the changes is doing the work, not which configuration to ship — for that see Section 14.

Both roles are produced by **trend** in every row, at the arrival tuning, because that is the parameter set being walked. The departure columns are therefore a diagnostic and not a recommendation: what ships takes departures from **endpoint**.

It was run twice — once with the pipeline’s original **H3 ring-count** selection, and once with **exact-distance** ranking. The two disagree in sign, which is the whole finding.

Step	Ring: score	Ring: gain	Exact: score	Exact: gain	Coverage	Accuracy
legacy constants	60,262	—	59,884	—	66.99%	98.00%
+ scheduled-service penalty	60,541	+279	60,709	+825	66.99%	98.43%
+ altitude ceiling	58,899	−1,642	61,500	+791	68.95%	97.93%
+ vote margin	59,023	+124	61,660	+160	69.14%	97.92%
+ radius	58,959	−64	61,467	−193	68.83%	97.96%
+ field-elevation datum	58,772	−187	61,478	+11	69.01%	97.89%

Under ring selection the tuning is worth $-1,490$; under exact distance it is worth $+1,594$. The altitude ceiling is the step that flips hardest — $-1,642$ against $+791$ — and it is the step that admits samples from higher up, where more aerodromes share a ring and the ring rule is least able to separate them.

The fifth rung is the datum, and it is the reason this version exists. Moving `trend`'s altitude cut from a flight level onto the aerodrome's own field elevation — at the ceiling admitting the identical band, so nothing else moves with it — is worth $+11$ under exact distance and -187 under ring selection. Section 6 takes that apart by field elevation, which is where the number stops being a single figure and starts being an explanation.

So the sweep harness was never over-estimating. It was measuring a *better algorithm* — one that measures every candidate before choosing — and the pipeline has now adopted it. `trend_rank_by` defaults to "haversine"; `DetectionConfig.legacy()` still returns "ring", so every published flight list stays reproducible.

12.2 The pipeline's own optimum

The path adopts the values the *sweep* preferred. Whether those are also the **pipeline's** optima is a separate question, and it is answered by sweeping the altitude ceiling and radius through `process_dai` directly — on the datum that ships, so the ceilings below are feet above field elevation.

Both roles are produced by `trend` in every cell of this grid. Held fixed throughout: vote margin 2, scheduled-service penalty 10 NM, ranking on exact distance. The vote margin is a `trend` parameter only — the majority one direction must win by across the smoothed altitude samples before a track is called a departure or an arrival. `endpoint` has no votes; it reads one fix.

The grid runs from 3,000 ft to 10,000 ft above field, which is far enough to find the arrival optimum and **not** far enough to close the departure side — see Section 21. It carries both 6,000 ft and 6,100 ft deliberately: the first is what ships and the second is what the research sweep preferred, and the sweep grid never contained the shipped value, so the two had never been compared until this table.

Score by altitude ceiling (rows) and radius in NM (columns), through the pipeline, at vote margin 2 and penalty 10 NM. Ceilings are heights above field elevation. Departures are shown for completeness – they are served by `endpoint`, so that half is a check rather than a recommendation.

Ceiling	ADES 20 NM	ADES 30 NM	ADEP 20 NM	ADEP 30 NM
3,000 ft above field	60,095	59,847	60,699	59,880
4,000 ft above field	61,093	60,743	64,182	63,429
6,000 ft above field	61,518	61,698	66,834	66,554
6,100 ft above field	61,478	61,654	66,898	66,635
8,000 ft above field	60,460	60,924	68,299	68,036
10,000 ft above field	58,242	58,873	69,637	69,335

Arrivals peak at **6,000 ft above field / 30 NM**. Departures climb to **10,000 ft above field**, but departures are served by `endpoint`, so that column is a check rather than a recommendation.

That arrival peak is interior, with the score falling away on both sides of the grid, so it is an optimum rather than the best cell that happened to be tried.

Which radius wins depends on the ceiling, and on the role. Departures prefer the tighter 20 NM at 6 of 6 ceilings; arrivals prefer it at only 2, and at every ceiling from 6,000 ft above field upward they prefer the wider 30 NM — which includes the ceiling that ships.

That matters because version 6 reported the opposite — that both roles prefer the tighter radius at every cell — and cited it as the stronger evidence for lowering `trend_radius_nm` to 20 NM. On the datum that ships, the grid does not support that for arrivals, which are the role `trend` actually serves. The value was never applied to `DetectionConfig` regardless, and this study does not renew the recommendation.

One caveat, stated rather than buried: the grid holds the **vote margin** fixed at 2 while varying the ceiling and radius. That value therefore rests on the sweep harness alone, and is the weakest-supported of the shipped parameters.

13 Why each value, and not its neighbour

A recommendation is only as good as the evidence that picked it, so every value the sweeps select appears below with the score at that value, the score at the adjacent grid points, and the score at the legacy value. The last column is the one that matters most for trusting the number: a value on a broad **plateau** is a safe recommendation even from a single sample, because its neighbours score nearly the same and a little noise cannot move it far. A value on a sharp isolated **peak** may be partly the sample’s noise, and is flagged as such however good its score.

! These are the sweeps’ choices, not the shipped ones

The **Chosen** column below is the **harness argmax**, which is what this section exists to scrutinise. For `trend`, Section 12 is what settles whether each value survives in the pipeline — and under exact-distance ranking they do, which is why the harness numbers below and the shipped values in Section 18 now agree. They are still kept apart, because for a period they did *not* agree, and a sweep preference is only ever evidence until the pipeline confirms it.

Role	Parameter	Legacy	Sweep argmax	Score at legacy	Score at argmax	Neighbour below	Neighbour above	Δ vs legacy	Shape
ADES	trend ceiling	4100	6100	—	61,056	60,814	59,999	—	plateau
ADES	trend vote margin	4	2	60,838	61,056	60,995	60,838	+218	plateau
ADES	trend radius (NM)	30	20	60,970	61,056	59,708	60,970	+86	plateau
ADEP	endpoint radius (NM)	40	30	71,674	71,881	71,754	71,674	+207	plateau
ADEP	endpoint height (ft)	15000	15000	71,881	71,881	70,785	71,523	+0	plateau

The penalty is treated separately because it is not a geometric parameter: it re-ranks candidates without changing how many are answered, so its sweep holds the geometry fixed and appears in Section 8.

13.1 Which change is actually doing the work

The table above varies one parameter at a time around the chosen setting, which shows whether each value is defensible but not how much each contributes to the total gain. This walks the whole distance instead: from the legacy constants to the recommended setting, changing one parameter per step.

How to read it. Each row *adds* its change to everything in the rows above and keeps it — row 5 has all four changes applied, not just the radius. **Gain** is that row’s score minus the previous row’s, so it is the marginal worth of that one change given everything already adopted. All rows are arrivals under **trend**, in the research sweep.

	Step	Score	Gain from this step	Share of total gain
legacy constants (4,000 ft above field, margin 4, 30 NM, no penalty)		56,322	—	—
+ scheduled-service penalty of 10 NM		59,900	+3,578	76%
+ altitude ceiling 6,100 ft above field		60,744	+844	18%
+ vote margin 2		60,970	+226	5%
+ radius 20 NM		61,056	+86	2%

The gain is not spread evenly. **scheduled-service penalty of 10 NM** accounts for 76% of it on its own. That is worth stating plainly: the single most valuable change to arrival detection is not a tuned threshold at all, but applying to **trend** the scheduled-service tie-break that **endpoint** already had and **trend** never did.

By contrast the radius step is worth 86 flights out of 61,056 here — inside what one three-day sample can resolve, and on this evidence alone it would not justify moving a published constant. Version 6 adopted it anyway, citing the pipeline grid as stronger evidence. On the datum that ships, that grid no longer supports it: see Section 12, where arrivals prefer the *wider* 30 NM at every ceiling from 6,000 ft up, which is where the shipped ceiling sits. The 20 NM recommendation was never applied to **DetectionConfig** in any case, and this study does not renew it.

14 The verdict

Configuration	ADEP	ADES	ADEP cover-age	ADEP accuracy	ADEP score	ADES cover-age	ADES accuracy	ADES score	Total
the same, built from DetectionConfig()	endpoint	trend	79.97%	98.21%	71,985	68.54%	98.44%	62,147	134,132
SHIPPED endpoint + exact-distance trend	endpoint	trend	79.98%	98.21%	71,981	67.68%	98.54%	61,556	133,537
endpoint, both roles	endpoint	endpoint	80.03%	98.14%	71,881	70.15%	96.22%	59,151	131,032
trend, departure parameters both roles	trend	trend	78.04%	97.92%	69,602	71.84%	95.67%	59,455	129,057

Configuration	ADEP	ADES	ADEP cover- age	ADEP accu- racy	ADEP score	ADES cover- age	ADES accu- racy	ADES score	Total
legacy — the algorithm published before 2026	trend	trend	67.46%	98.82%	61,896	66.99%	98.19%	60,262	122,158
nearest (traffic library)	nearest	nearest	91.64%	87.68%	54,946	88.55%	79.29%	31,891	86,837

Each row is a complete configuration — which method names departures, which names arrivals — not a step in a sequence. The rows are alternatives to choose between.

The recommended configuration is the **combined** row: **endpoint** for departures, exact-distance **trend** for arrivals. It scores **133,537** against legacy’s **122,158**, a gain of **+11,379** on a three-day sample of 95,116 flights, and every figure in that table is a real pipeline run scored against Network Manager ground truth.

It is the shipped configuration in the strict sense: its trend cap, radius, vote margin, penalty and ranking rule, and its endpoint radius, height and penalty, were checked field by field against `DetectionConfig()`. The row is what the pipeline produces, not an approximation of it.

The **recommended** row is the same configuration constructed the other way — by instantiating `DetectionConfig()` rather than by naming each field — and it lands within 595 of **combined** in total. Two independent constructions of the shipped settings agreeing to that margin is the check that the recommendation and the code have not drifted apart.

14.1 What is being recommended, and why

Departures — endpoint, 30 NM radius, 15,000 ft above field elevation, 10 NM scheduled-service penalty. That height is `endpoint_height_ft`, a height above the aerodrome’s own elevation — not a flight level, and not related to the flight-level caps swept for **trend**, which departures do not use. Score 71,981 against legacy’s 61,896, a gain of +10,085. The clearest result in the study: an interior optimum in both parameters, better than **trend** at every exchange rate tried, and the pipeline reproduces the sweep exactly — the endpoint sweep filters the same cached candidate table the pipeline reads, so the tuned figure *is* the pipeline figure.

Arrivals — trend at 6,000 ft above field, vote margin 2, 30 NM, 10 NM penalty, ranked on exact distance. Measured as an algorithm, on a trend-only flight list, that is 61,698 against legacy’s 60,262 — a gain of +1,436, won by converting 1,968 silences into correct answers for 266 extra wrong ones.

Inside the combined flight list the arrival half reads 61,556, +1,294 against legacy rather than the +1,436 just quoted. The difference is not the algorithm. Taking departures from **endpoint** adds tracks **trend** never produced, and where two share an aircraft, callsign and date, the scoring alignment awards the flight to whichever starts closest to the actual off-block time — sometimes now a departure-only track carrying no arrival. Accuracy *rises* in the merged table (98.54% against 98.19%) while coverage falls, which is the signature of lost pairings rather than wrong names.

One weak link, stated plainly. The vote margin of 2 comes from the sweep harness; the pipeline grid held it fixed while varying the cap and radius. Every other shipped value was

measured through `process_dai`. If one parameter in this recommendation is going to fail on a fresh sample, that is the one.

14.2 The traffic library's rule, for reference

`nearest` — the unconditional rule the `traffic` library uses — answers far more often than anything else here (91.64% of departures, 88.55% of arrivals) and is right far less often (87.68% and 79.29%). Scored, it is last by a wide margin: 86,837 against the recommendation's 134,132.

That is not a criticism of `traffic`, which is solving a different problem: it names the nearest aerodrome and leaves the judgement to the caller. It is, however, the measurement that justifies OPDI carrying its own rule rather than adopting that one.

15 By aerodrome class

The candidate set is large and medium aerodromes only, so a flight whose true aerodrome is a small airfield or a heliport cannot be named correctly by any method on this page. This is the size of that population, and of everything else.

Departures

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,661	331	70,353	69,786	87.22%	99.19%
out of area	7,892	1	921	628	11.67%	68.19%
medium_airport	6,069	352	4,634	4,293	76.36%	92.64%
small_airport	361	76	134	0	37.12%	0.00%
heliport	133	65	28	0	21.05%	0.00%

Arrivals

Class	Flights in truth	Aerodromes	Answered	Correct	Coverage	Accuracy
large_airport	80,683	331	60,528	60,145	75.02%	99.37%
out of area	7,964	1	17	0	0.21%	0.00%
medium_airport	5,983	346	3,738	3,291	62.48%	88.04%
small_airport	352	70	83	0	23.58%	0.00%
heliport	134	58	10	0	7.46%	0.00%

Outside the candidate set — 3 classes, 16,836 flights across both roles — the method is working against a reference it was never given. Those flights are counted in the denominator throughout this page, so every coverage figure here already carries their cost.

16 The busiest 100 aerodromes

Movement counts are what OPDI is for, so the last check is per aerodrome rather than per flight: for each of the hundred busiest in the reference, how many movements the recommended methodology finds against how many the Network Manager data says there were.

Across those hundred, the shipped configuration recovers **79.48%** of arrivals and **91.52%** of departures, against the legacy algorithm's 79.11% and 82.05% on the same aerodromes. The aerodrome-by-aerodrome tables are in Section 22.

17 Does any of it hold on a second period?

Everything above is one three-day sample. An argmax over hundreds of cells on a single sample always flatters itself: part of the winning cell's margin is that sample's noise, and noise does not reproduce. The only way to find out which part is real is to measure a different period with the same code.

The second period is **5–7 June 2024**, chosen because the Network Manager reference for that month is committed to the repository and the matching tracks already exist from earlier versions of this study.

The 2024 sample carries 92,799 ground-truth flights against 2025's 95,116.

Role	Parameter	2025 optimum	2024 optimum	Agrees
ADEP	trend ceiling	12000	10000	no
ADEP	trend vote margin	0	0	yes
ADEP	trend radius (NM)	20	20	yes
ADES	trend ceiling	6100	6100	yes
ADES	trend vote margin	2	0	no
ADES	trend radius (NM)	20	20	yes

4 of 6 parameter optima agree across the two periods — which, taken alone, reads worse than the situation is. Comparing two argmaxes is close to the weakest test available: over hundreds of cells the top cell moves under noise even when the surface underneath is stable.

So the cells are ranked here by **both periods at once**, with each period's score divided by its own ground-truth count so the larger sample cannot dominate. On that ranking the shipped arrival configuration — 6,100 ft above field, vote margin 2, 20 NM, 10 NM penalty — places **1 of 280**. The joint winner is 6,100 ft above field / margin 2 / 20 NM, and differs by 0.00000 of normalised score.

One substitution, stated rather than buried: the shipped ceiling is 6,000 ft above field, and the research sweep's grid does not contain it — the grid was built around 6,100 ft above field, the height equivalent of FL60. So 6,100 ft above field stands in above, and the ranking is that cell's rank, not the shipped cell's. The pipeline grid in Section 12 carries both values and is where the two are actually compared.

The arrival tuning therefore does hold on both periods. What moves between them is which of several near-equivalent cells happens to come first, and the radius is the parameter that moves.

For the endpoint rule, which is what departures actually adopt:

Role	2025 radius / height	2024 radius / height	Agrees
ADEP	30 NM / 15,000 ft	30 NM / 15,000 ft	yes
ADES	30 NM / 10,000 ft	30 NM / 10,000 ft	yes

The departure optimum — the one that actually ships — is the *same cell* on two independent periods, out of 126. Neither period was consulted when the other was measured. That is the strongest stability evidence in this study.

18 What ships

This table is **read from the runs themselves**, not written out here. The **legacy** column is the configuration the **legacy** run was built with, and the **ships** column is the configuration the **recommended** run was built with — and **recommended** is constructed as a bare `DetectionConfig()`, so it is by definition whatever the pipeline currently defaults to.

That matters more than it sounds. The equivalent table in version 6 was typed by hand, and it had drifted: it named two values the code does not have. A table asserting what ships, on a page whose whole argument is that hand-copied figures cannot be trusted, is the one place that must not be hand-copied.

Parameter	Legacy	Ships	Evidence
<code>trend_rank_by</code>	<code>haversine</code>	<code>file</code>	The enabling change. Ring-count selection discarded candidates before measuring them; every tuned trend value fails under it and gains under exact distance.
<code>trend_max_datum</code>	<code>msl</code>	<code>field</code>	Section 6. Aligns <code>trend</code> 's altitude cut with every other altitude gate in the pipeline.
<code>trend_max_height</code>	<code>6,000</code>	<code>6,100</code>	The ceiling that datum is measured against, and inert while the datum is <code>msl</code> — which is why the legacy column shows a value legacy never applied. See Section 12 on 6,000 against 6,100.
<code>trend_max_f40</code>	<code>60</code>	<code>60</code>	Retained, and inert while the datum is <code>field</code> . It is what <code>legacy()</code> reads, so published lists stay reproducible.
<code>trend_radius</code>	<code>30 nm</code>	<code>30 nm</code>	Swept in Section 5 and through the pipeline in Section 12.
<code>trend_vote_margin</code>	<code>4</code>	<code>0</code>	Sweep harness only — the pipeline grid holds it fixed. The weakest-supported value here.
<code>trend_sched_penalty</code>	<code>10</code>	<code>1</code>	A tie-break <code>trend</code> never had. The single largest contributor to the arrival gain.
<code>endpoint_radius</code>	<code>30 nm</code>	<code>30 nm</code>	Interior optimum; the pipeline reproduces the sweep exactly.
<code>endpoint_height</code>	<code>5,000</code>	<code>5,000</code>	A height above field elevation, and always was.
<code>endpoint_sched_penalty</code>	<code>1</code>	<code>1</code>	Unchanged.

6 of the 10 parameters differ from the legacy constants.

! What ships is not what the sweeps chose

The shipped configuration and this study's swept optimum differ in 3 parameter(s):

Parameter	Ships	Sweep optimum
<code>trend_radius_nm</code>	30	20
<code>trend_vote_margin</code>	0	2
<code>trend_ceiling</code>	6000	6100

Both are measured above: **recommended** is the shipped row, **combined** the swept one, and on arrivals they score 62,147 against 61,556. This is reported rather than resolved — changing a published default is a decision for the pipeline’s maintainers, and this page’s job is to make the gap visible.

6,000 ft or 6,100? The shipped ceiling is 6,000 ft; the research sweep called 6,100 optimal — on a grid that never contained 6,000, so that recommendation was never a comparison. Run through `process_dai` itself, the best arrival score at 6,000 ft is 61,698 and at 6,100 ft is 61,654, a difference of -44 . That is inside what one three-day sample resolves, so the shipped 6,000 ft stands: there is no measured reason to move it, and moving a published constant needs one.

The ceiling and the vote margin were **rejected earlier in this study and reinstated**. That rejection was measured under ring-count selection, where raising the ceiling genuinely did lose ground. The lesson is not that the first measurement was careless but that it answered a narrower question than it appeared to: *given this selection rule*, the tuning fails. Changing the rule changed the answer.

The radius is a different story, and version 6 told it wrongly. Its table recorded `trend_radius_nm` as shipping at 20 NM. The table above — read from a bare `DetectionConfig()` rather than typed — shows it at **30 NM**, unchanged from legacy. The recommendation was made and never applied, and no later version noticed because every later version copied the claim rather than measuring it.

Every OPDI flight list published before this change was built with the legacy column, and `DetectionConfig.legacy()` still returns exactly those values — including `trend_rank_by = "ring"` — so anything already released stays reproducible by name. That is the same principle as the frozen **version** strings on published events. A regression test pins the preset, a second pins the shipped values, and a third asserts the two differ, so the preset cannot quietly become vacuous.

⚠ Running it

The configuration this report recommends is the **default**. `adep_mode` and `ades_mode` default to `None`, which means the pipeline uses `DetectionConfig`’s own values — so you get the recommendation by not overriding anything.

Step 03 is the flight list: one row per track, carrying ADEP and ADES.

```
from datetime import date
from opdi.runner import run_pipeline

run_pipeline(
    env="opensky",                # plain parquet over S3A; no Hive, no Iceberg
    step="03",
    start_date=date(2025, 6, 1),
    end_date=date(2025, 6, 30),
)
```

The same thing from a shell, using the installed console script:

```
opdi run --env opensky --step 03 --start 2025-06-01 --end 2025-06-30
```

To depart from the recommendation, name the methods explicitly. Both accept **trend**, **endpoint** or **nearest**:

```
opdi run --env opensky --step 03 --start 2025-06-01 --end 2025-06-30 \  
--adep-mode trend --ades-mode trend
```

That example asks for **trend** on both roles — the pre-2026 behaviour, and the **legacy** row in every comparison on this page.

Step 03 does not stand alone. It reads the tracks written by step 02 and the aerodrome zones from step 00, so those must have run for the same period first. Omitting **--step** runs the whole chain in order.

What “recommended” means here, and how it is kept honest

The values are not written into this page. Section 18 reads them from a bare `DetectionConfig()` — the same object the pipeline constructs — so the table cannot describe a configuration the code does not have. The equivalent table in version 6 was typed by hand and named two values the code never had, for three versions.

`benchmarks/check_main_parity.py` closes the loop the other way: it compares that configuration against the best cell of the pipeline grid, which is the only sweep that runs the shipped package rather than a re-implementation of it.

19 Re-running a past month will not reproduce it

These defaults change ADEP and ADES for any month re-processed with them. That is intended, and it is why the old behaviour remains reachable by name rather than being deleted. A new algorithm gets a new **version**; the published one is never mutated.

20 Provenance

Each row is one file this page reads: the job that produced it, the commit it was produced at, and when. Every figure on this page is regenerated by `benchmarks/regenerate_v62.py`, which the page runs before drawing anything, so a row marked **unverified** would mean a file staged outside that chain and traceable to no command.

File	Job	Commit	Produced
bearing_whole_sample_v6.csv	bearing_whole_sample.py	6a3121e	2026-08-24T10:59
datum_swap_2025.csv	flight_list_v62.py	56cc9b5	2026-08-24T21:10
elevation_bands_2025.csv	elevation_arms.py	96805df (dirty)	2026-08-24T21:14

File	Job	Commit	Produced
elevation_per_airport_2025.csv	elevation_arms.py	96805df (dirty)	2026-08-24T21:14
fl_sweep_2024.csv	trend_sweep_agl.py	53bd618	2026-08-23T18:04
fl_sweep_2025.csv	trend_sweep_agl.py	dd918d9	2026-08-23T16:51
height_sweep_2024.csv	trend_sweep_agl.py	53bd618	2026-08-23T17:53
height_sweep_2025.csv	trend_sweep_agl.py	dd918d9	2026-08-23T16:40
input_drift.csv	input_drift.py	5ced37c (dirty)	2026-08-24T15:02
merge_agreement_v6.csv	merge_diagnosis.py	6a3121e	2026-08-24T11:02
merge_shapes_v6.csv	merge_diagnosis.py	6a3121e	2026-08-24T11:02
mode_comparison_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T18:43
per_airport_datum_2025.csv	flight_list_v62.py	56cc9b5	2026-08-24T21:10
per_airport_path_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T20:27
per_airport_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T18:43
per_type_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T18:43
pipeline_path_ring_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T20:12
pipeline_path_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T20:27
sampler_comparison_v6.csv	sampler_comparison.py	f0f68e4 (dirty)	2026-08-11T08:16
sweep_cone_2025.csv	benchmark_modes.py	6a3121e	2026-08-24T10:51
sweep_equivalence.csv	sweep_equivalence.py	329eb8c (dirty)	2026-08-24T09:28
sweep_penalty_2025.csv	benchmark_modes.py	6a3121e	2026-08-24T10:51
sweep_radius_height_2024.csv	benchmark_modes.py	6a3121e	2026-08-24T10:56
sweep_radius_height_2025.csv	benchmark_modes.py	6a3121e	2026-08-24T10:51
trend_bearing_v6.csv	trend_bearing.py	6a3121e	2026-08-24T11:04
trend_grid_v6.csv	flight_list_v62.py	56cc9b5	2026-08-24T19:40
vertical_measure_v6.csv	vertical_measure.py	6a3121e	2026-08-24T11:01

27 files, 27 traced to a command, 0 unverified.

20.1 Where reproducibility stops

Two boundaries, stated rather than implied.

Ground truth cannot be rebuilt here. `research/reference` is extracted by the `eurocontrol` R package against the PRISME Oracle warehouse, which no run on this cluster can reach. It is an external input, and every figure on this page is measured against it.

The upstream tables are rebuilt only when absent. Reference zones, state vectors and tracks are produced by declared stages, but a stage whose table is already present records its identity rather than rebuilding it — because regenerating the airport zone table would change the candidate set under the whole study. For this version they *were* rebuilt, from the archive upward, because the sampler changed.

20.2 What moved underneath this version

This version sets out to recompute version 6’s study with **one** variable changed. That is a checkable claim, and it does not fully hold. Stating where it fails is the point of recording input identities in the first place.

Table	Objects	Size	Newest	Verdict
<code>h3_airport_detection_zones</code>	152 → 151	0.22 → 0.22 GB	2026-08-07 → 2026-08-07	marker only
<code>opdi_endpoint_candidates</code>	24 → 23	0.17 → 0.15 GB	2026-08-10 → 2026-08-22	REBUILT
<code>osn_statevectors_v2</code>	202 → 0	9.68 → 0.00 GB	2026-08-10 → —	table now absent or empty
<code>osn_tracks</code>	5 → 4	10.16 → 10.16 GB	2026-08-10 → 2026-08-10	marker only
<code>research/cand_2024</code>	24 → 23	0.13 → 0.12 GB	2026-08-10 → 2026-08-12	REBUILT
<code>research/reference</code>	4 → 6	0.28 → 0.54 GB	2026-08-05 → 2026-08-13	REBUILT
<code>research/trend_votes</code>	302 → 301	0.45 → 0.62 GB	2026-08-10 → 2026-08-12	REBUILT
<code>research/trend_votes_2024</code>	302 → 301	0.41 → 0.58 GB	2026-08-10 → 2026-08-12	REBUILT

`h3_airport_detection_zones`, `osn_tracks`: byte-identical, same timestamp, one object fewer. That is the zero-byte directory marker MinIO leaves behind, which `provenance.s3_identity` stopped counting after it made a *deleted* table look populated. Bookkeeping, with no consequence for any figure.

`opdi_endpoint_candidates`, `research/cand_2024`, `research/reference`, `research/trend_votes`, `research/trend_votes_2024`: genuinely rebuilt since version 6 recorded them. Every figure derived from them moved, and the datum does not explain any of it.

The one that matters is `opdi_endpoint_candidates`, rebuilt after version 6 published: about 12% smaller, with the fresh-broadcast share roughly tripled — the signature of stale-broadcast removal, which the PRC challenge results recommend and which `endpoint` reads directly.

Two consequences, and they pull in opposite directions.

The ground truth did *not* move. `research/reference` grew by more than half, but it grew by gaining *other periods*: this study’s three-day slice is unchanged, at 95,116 reference flights on 2025 and 92,799 on 2024 — the same counts version 6 reported. That is confirmed independently by Section 6.1, where the sea-level sweep reproduces version 6’s cell for cell. So every denominator on this page is the one version 6 used, and the internal comparisons — which are what this study actually argues from — are sound.

But the endpoint-family numbers are not comparable to version 6’s. The endpoint sweeps, the bearing test and the vertical measures all read the rebuilt candidate table. Their figures differ from version 6’s, and that difference is a data change rather than a finding.

 A verification this version cannot make

The intention was to verify directly that the datum change leaves departures alone: the endpoint family is datum-independent by construction, so recomputing it under the new code and recovering version 6’s numbers would have settled it.

The candidate rebuild makes that comparison uncontrolled, so the verification is **not available** and is not claimed. What remains is the within-study evidence in Section 6.2 — departures gain nothing in *any* elevation band, measured where both arms read the same candidate table. That is genuine evidence and it is weaker than the check that was intended.

21 Limitations

21.1 Parameters that were not swept

Five parameters were held fixed by decision rather than by evidence. Naming them matters: a reader who sees seven tuned values and no caveat will reasonably assume the whole configuration was optimised, and it was not.

Parameter	Where	Held at	Why not swept
Smoothing half-window	<code>flights.py</code>	2 samples	Parameterised in config so it is visible and changeable, but a smoothing width interacts with the sample rate rather than the geometry, and the decimation study settled the sample rate separately.

Parameter	Where	Held at	Why not swept
Classification dead band	<code>flights.py</code>	none exists	There is no dead band to sweep. The vote margin already plays that role.
Out-of-area border margin	<code>flights.py</code>	30 NM	Fixes what counts as “outside the observed area”. Changing it redefines the ground truth rather than the method, so a sweep would move the target as well as the aim.
Aerodrome type set	<code>h3_airport_zones.py</code>	large + medium	The candidate set. Widening it is a data-volume decision about the zone table, not a detection threshold.
H3 resolution	<code>config.py</code>	7 (± 0.76 NM)	Changing it invalidates every cached zone table. It no longer decides which candidates survive — that is exact distance now — but it still decides which aerodromes become candidates at all.

Two of these carry a cost that this study measures rather than waves at, since “not swept” is not the same as “no consequence”.

The type set. 980 flights across both roles have a true aerodrome outside the large/medium candidate set (267 at heliports, 713 at small airports). No method on this page can name them, whatever its thresholds, because the aerodrome is not among the candidates. They are counted in every denominator here, so each coverage figure already carries their cost: they are worth about 0.52% of headline coverage.

The border margin. Fixed at 30 NM, it decides which flights are labelled out-of-area. 8.30% of ground-truth departures genuinely begin outside the observed area, and the recommended configuration recognises 7.96% of them — at 89.20% precision, so the label is trustworthy when it appears but is rarely reached. For arrivals the recall is 0.00%: `trend` emits no out-of-area label at all, which is a structural property of the vote rule rather than a tuning choice.

Both costs fall on coverage rather than accuracy, which is the benign direction: with the free out-of-area wins removed, in-area accuracy is 98.58% for departures and 98.48% for arrivals.

21.2 trend cannot say “out of area”

`trend` names an aerodrome or says nothing. It has no way to report that a flight began or ended outside the observed area, because the vote rule only ever considers aerodromes the aircraft flew near — silence and *outside* are the same output.

That is not a small omission for the shipped configuration, which takes arrivals from `trend`:

8.37% of reference arrivals genuinely come from outside the observed area, and **none of them can be labelled as such**. Departures ship `endpoint`, which does emit the label: 89.20% precision, so it is trustworthy when it appears, but only 7.96% recall.

`endpoint` already has the mechanism — a fix at the edge of the ingestion region rather than near any aerodrome — so giving `trend` the same test is a port, not an invention. It is the clearest single improvement left on this page and is not implemented here.

21.3 The grid stops before the departure optimum does

The pipeline grid runs to 10,000 ft above field, and departures under `trend` are still rising there. The research sweep goes further and does turn — departures peak near the top of its range on both periods, then fall away as accuracy drops faster than coverage rises — but “it turns in the sweep” is weaker evidence than “it turns in the pipeline”, and the two have disagreed before.

This costs nothing that ships, because departures are served by `endpoint` and the arrival optimum is comfortably interior on both periods and in both stages. It does mean the departure half of the trend grid should not be quoted as an optimum.

21.4 One sample, and what that does and does not license

The parameters are chosen as the argmax over a grid evaluated on a single three-day period. An argmax over hundreds of cells on one sample will always overstate the gain slightly: some of the winning cell’s margin is the sample’s noise, and it will not reproduce.

That is the reason for the second period, and for reporting the shape of each optimum rather than only its location. A parameter sitting on a broad plateau is a safe recommendation even from one sample; one sitting on a sharp isolated peak is not, however good its score.

21.5 The second period confirms through the sweeps, not the pipeline

Every pipeline number on this page is 2025-06-05...07. The 2024 tracks pre-date this study’s H3 indexing and `process_dai` reads `h3_res_7` off the track table directly, so the pipeline arms cannot run on that period at all — version 6 met the same wall and ran the pipeline on 2025 alone. Section 6.6 states the consequence where it arises.

Repeated here because it bounds every two-period claim in this report: the second period tests the ceiling and the datum comparison, which is the part most at risk of being a one-sample artefact, and it does **not** independently reproduce the band breakdown, the tuning ladder, or the pipeline fidelity check. A caveat that appears only beside the result it qualifies is easy to read past.

21.6 The >3000 band needs a different question

The highest elevation band is not a datum problem and this study cannot fix it. Recall there is under one percent on **both** datums, which is a coverage failure upstream of detection: the aerodrome never becomes a candidate, so no threshold anywhere in **trend** can reach it. Whether that is receiver geometry over the Anatolian plateau or something in ingestion is worth knowing and is not this study’s question.

It is worth stating plainly because that band is where an elevation-aware cut should have helped most, and the fact that it did not is evidence about coverage rather than evidence against the datum — Section 6.2.

22 Appendix A — The busiest 100 aerodromes

Movement counts are what OPDI is for, so the last check is per aerodrome rather than per flight: for each of the hundred busiest in the reference, how many movements the recommended methodology finds, against how many the Network Manager data says there were, and how the legacy algorithm compares on the same aerodromes.

The matching is **flight by flight, not count against count**. Each reference flight is paired with an OPDI track on aircraft address (**icao24**) plus callsign plus date, and where several tracks match, the one starting closest to the reference off-block time. The columns then mean:

Column	Definition
Movements (truth)	Reference flights whose true aerodrome is this one.
Found	Of those, how many OPDI gave any answer for.
Correct	Of those, how many OPDI attributed to this aerodrome.
Count ratio	Movements OPDI <i>assigns to</i> this aerodrome ÷ movements in truth. Includes flights that truly belong elsewhere, so it is what an operational count would show, and it can exceed 1.
Recall	Correct ÷ movements in truth. Per-aerodrome recall — not the out-of-area recall of Section 2, which is a different measure sharing the word.
Recall, legacy	The same, for the algorithm published before 2026.

22.1 Departures

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
1	LTFM	2,282	2,173	2,166	0.953	94.92%	57.45%	+37.47 pp
2	EHAM	2,148	2,143	2,143	1.005	99.77%	99.81%	-0.05 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
3	EDDF	2,094	2,080	2,080	0.994	99.33%	96.94%	+2.39 pp
4	LFPG	2,084	2,065	2,061	0.994	98.90%	99.04%	-0.14 pp
5	EGLL	2,001	1,989	1,988	1.011	99.35%	99.30%	+0.05 pp
6	LEMD	1,773	1,758	1,757	0.991	99.10%	93.97%	+5.13 pp
7	LEBL	1,591	1,578	1,577	0.991	99.12%	99.43%	-0.31 pp
8	EDDM	1,573	1,561	1,561	0.994	99.24%	99.17%	+0.06 pp
9	LIRF	1,449	1,442	1,442	0.995	99.52%	99.72%	-0.21 pp
10	LTAI	1,435	1,229	1,228	0.860	85.57%	50.66%	+34.91 pp
11	LEPA	1,417	1,360	1,360	0.960	95.98%	87.37%	+8.61 pp
12	LGAV	1,306	1,238	1,238	0.956	94.79%	23.05%	+71.75 pp
13	EGKK	1,223	1,213	1,213	0.993	99.18%	99.59%	-0.41 pp
14	EIDW	1,185	1,164	1,163	0.981	98.14%	98.23%	-0.08 pp
15	LSZH	1,178	1,170	1,163	0.990	98.73%	98.47%	+0.25 pp
16	LOWW	1,156	1,137	1,136	0.990	98.27%	98.88%	-0.61 pp
17	EKCH	1,140	1,132	1,132	0.996	99.30%	97.98%	+1.32 pp
18	LTFJ	1,137	1,092	1,091	0.972	95.95%	90.59%	+5.36 pp
19	LPPT	1,010	1,001	1,001	0.991	99.11%	79.50%	+19.60 pp
20	LFPO	992	982	982	0.997	98.99%	99.29%	-0.30 pp
21	LIMC	973	964	963	0.991	98.97%	99.28%	-0.31 pp
22	ENGM	932	919	919	0.991	98.61%	86.48%	+12.12 pp
23	EGCC	926	906	905	0.977	97.73%	99.14%	-1.40 pp
24	EGSS	885	880	880	0.994	99.44%	99.32%	+0.11 pp
25	EPWA	875	869	868	0.997	99.20%	99.43%	-0.23 pp
26	EBBR	872	866	866	1.006	99.31%	99.54%	-0.23 pp
27	EDDB	870	852	850	0.993	97.70%	97.47%	+0.23 pp
28	LEMG	850	846	846	0.998	99.53%	99.53%	+0.00 pp
29	ESSA	820	771	767	0.940	93.54%	96.34%	-2.80 pp
30	EDDL	758	747	745	0.989	98.28%	98.94%	-0.66 pp
31	LFMN	707	697	691	0.979	97.74%	97.74%	+0.00 pp
32	LSGG	680	675	675	1.009	99.26%	65.44%	+33.82 pp
33	LKPR	667	657	656	0.990	98.35%	98.65%	-0.30 pp
34	EFHK	659	653	653	1.002	99.09%	99.24%	-0.15 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
35	LEAL	599	597	597	0.997	99.67%	99.83%	-0.17 pp
36	LHBP	578	555	553	0.958	95.67%	97.92%	-2.25 pp
37	EGPH	577	571	570	0.988	98.79%	98.96%	-0.17 pp
38	EGGW	550	549	549	1.002	99.82%	97.82%	+2.00 pp
39	EDDH	538	531	528	0.998	98.14%	98.33%	-0.19 pp
40	GCLP	520	23	0	0.000	0.00%	0.00%	+0.00 pp
41	EDDK	515	500	500	0.973	97.09%	96.89%	+0.19 pp
42	LLBG	513	500	500	0.994	97.47%	96.69%	+0.78 pp
43	LPPR	513	511	508	0.990	99.03%	99.03%	+0.00 pp
44	LROP	504	491	491	0.974	97.42%	98.21%	-0.79 pp
45	LEIB	495	484	484	0.978	97.78%	84.44%	+13.33 pp
46	EGBB	474	468	468	0.987	98.73%	98.52%	+0.21 pp
47	LIML	458	451	451	0.987	98.47%	98.47%	+0.00 pp
48	LIME	453	450	450	0.993	99.34%	98.45%	+0.88 pp
49	LIRN	449	380	334	0.751	74.39%	0.00%	+74.39 pp
50	LFML	444	441	440	0.991	99.10%	98.87%	+0.23 pp
51	EDDS	440	420	414	0.948	94.09%	94.32%	-0.23 pp
52	LGIR	424	6	0	0.000	0.00%	0.00%	+0.00 pp
53	LIPZ	412	408	408	0.993	99.03%	98.79%	+0.24 pp
54	LFLL	411	402	402	1.015	97.81%	98.78%	-0.97 pp
55	LPFR	405	400	398	0.983	98.27%	98.77%	-0.49 pp
56	LYBE	394	388	388	0.987	98.48%	98.22%	+0.25 pp
57	LICC	390	352	348	0.892	89.23%	0.00%	+89.23 pp
58	LTAC	381	7	0	0.000	0.00%	0.00%	+0.00 pp
59	LIPE	380	376	376	0.989	98.95%	98.42%	+0.53 pp
60	LCLK	368	353	352	0.962	95.65%	85.87%	+9.78 pp
61	GMMN	360	287	282	0.986	78.33%	40.83%	+37.50 pp
62	EPKK	360	348	346	0.964	96.11%	97.50%	-1.39 pp
63	EGGD	359	355	354	0.986	98.61%	99.44%	-0.84 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
64	LEVC	355	337	336	0.946	94.65%	89.58%	+5.07 pp
65	LFSB	354	351	349	0.994	98.59%	98.31%	+0.28 pp
66	ENBR	349	343	343	1.026	98.28%	96.56%	+1.72 pp
67	LTBJ	346	274	229	0.662	66.18%	0.29%	+65.90 pp
68	EGPF	344	325	323	0.951	93.90%	93.60%	+0.29 pp
69	LGRP	340	322	321	0.962	94.41%	93.53%	+0.88 pp
70	LMML	326	325	325	1.000	99.69%	98.77%	+0.92 pp
71	GCTS	321	4	0	0.019	0.00%	0.00%	+0.00 pp
72	GCXO	319	305	305	1.135	95.61%	20.38%	+75.24 pp
73	LATI	316	6	0	0.000	0.00%	0.00%	+0.00 pp
74	LICJ	310	281	281	0.910	90.65%	44.84%	+45.81 pp
75	EBCI	304	303	303	0.997	99.67%	100.00%	-0.33 pp
76	BIKF	297	296	294	1.084	98.99%	97.98%	+1.01 pp
77	ELLX	294	291	291	0.990	98.98%	98.98%	+0.00 pp
78	LFBO	293	288	288	0.990	98.29%	98.98%	-0.68 pp
79	LGTS	279	260	259	0.928	92.83%	41.58%	+51.25 pp
80	GMMX	278	74	66	0.241	23.74%	0.36%	+23.38 pp
81	LEZL	275	275	274	0.996	99.64%	99.64%	+0.00 pp
82	LTBS	273	205	205	0.751	75.09%	0.00%	+75.09 pp
83	EVRA	273	270	269	0.989	98.53%	98.53%	+0.00 pp
84	LBSF	272	153	153	0.562	56.25%	0.00%	+56.25 pp
85	GCRR	269	1	0	0.000	0.00%	0.00%	+0.00 pp
86	EGNX	267	263	263	0.985	98.50%	98.88%	-0.37 pp
87	EPKT	259	251	249	0.961	96.14%	92.66%	+3.47 pp
88	HEGN	257	0	0	0.000	0.00%	0.00%	+0.00 pp
89	UGTB	257	157	151	0.747	58.75%	0.00%	+58.75 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
90	LFRS	256	247	247	0.969	96.48%	71.88%	+24.61 pp
91	EDDP	256	255	255	1.004	99.61%	98.05%	+1.56 pp
92	EPGD	248	35	17	0.069	6.85%	2.42%	+4.44 pp
93	LGKR	244	203	201	0.844	82.38%	0.00%	+82.38 pp
94	HECA	244	48	30	0.123	12.30%	1.64%	+10.66 pp
95	EGAA	239	232	232	0.975	97.07%	61.92%	+35.15 pp
96	ENZV	236	151	148	0.686	62.71%	29.24%	+33.47 pp
97	LEBB	234	180	169	0.731	72.22%	0.00%	+72.22 pp
98	LFPB	230	227	227	1.000	98.70%	95.65%	+3.04 pp
99	LIBD	228	1	0	0.000	0.00%	0.00%	+0.00 pp
100	LIEO	228	182	175	0.768	76.75%	3.95%	+72.81 pp

22.2 Arrivals

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
1	LTFM	2,258	1,822	1,772	0.785	78.48%	70.50%	+7.97 pp
2	EHAM	2,136	1,876	1,875	0.879	87.78%	90.17%	-2.39 pp
3	EDDF	2,091	1,775	1,772	0.848	84.74%	87.61%	-2.87 pp
4	LFPG	2,066	1,715	1,714	0.831	82.96%	85.82%	-2.86 pp
5	EGLL	1,986	1,582	1,580	0.852	79.56%	85.45%	-5.89 pp
6	LEMD	1,786	1,536	1,533	0.861	85.83%	89.98%	-4.14 pp
7	LEBL	1,597	1,497	1,494	0.936	93.55%	95.68%	-2.13 pp
8	EDDM	1,571	1,468	1,465	0.936	93.25%	94.40%	-1.15 pp
9	LTAI	1,437	855	855	0.597	59.50%	52.96%	+6.54 pp
10	LEPA	1,436	1,413	1,412	0.985	98.33%	96.31%	+2.02 pp
11	LIRF	1,433	1,256	1,253	0.875	87.44%	90.16%	-2.72 pp
12	LGAV	1,317	530	524	0.399	39.79%	8.88%	+30.90 pp
13	EGKK	1,220	1,103	1,103	0.904	90.41%	92.54%	-2.13 pp
14	EIDW	1,188	1,084	1,082	0.912	91.08%	93.35%	-2.27 pp
15	LSZH	1,160	1,053	1,049	0.910	90.43%	92.07%	-1.64 pp
16	EKCH	1,152	1,045	1,045	0.907	90.71%	91.84%	-1.13 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
17	LOWW	1,145	1,045	1,042	0.912	91.00%	93.62%	-2.62 pp
18	LTFJ	1,138	990	984	0.874	86.47%	87.61%	-1.14 pp
19	LPPT	1,001	881	881	0.880	88.01%	90.51%	-2.50 pp
20	LFPO	991	941	941	0.951	94.95%	95.76%	-0.81 pp
21	LIMC	967	856	852	0.881	88.11%	91.52%	-3.41 pp
22	ENGM	940	889	889	0.952	94.57%	94.15%	+0.43 pp
23	EGCC	924	819	818	0.885	88.53%	90.91%	-2.38 pp
24	EGSS	888	832	831	0.938	93.58%	94.71%	-1.13 pp
25	EPWA	880	752	751	0.853	85.34%	88.64%	-3.30 pp
26	EBBR	871	767	767	0.890	88.06%	89.90%	-1.84 pp
27	LEMG	861	834	834	0.969	96.86%	97.91%	-1.05 pp
28	ESSA	841	726	721	0.863	85.73%	86.09%	-0.36 pp
29	EDDB	840	789	788	0.948	93.81%	94.64%	-0.83 pp
30	EDDL	751	717	715	0.959	95.21%	97.20%	-2.00 pp
31	LFMN	710	684	676	1.038	95.21%	96.62%	-1.41 pp
32	LSGG	670	637	637	0.963	95.07%	96.72%	-1.64 pp
33	EFHK	666	572	572	0.862	85.89%	86.49%	-0.60 pp
34	LKPR	657	605	605	0.928	92.09%	93.76%	-1.67 pp
35	LEAL	599	588	588	0.982	98.16%	98.83%	-0.67 pp
36	EGPH	584	541	541	0.933	92.64%	94.69%	-2.05 pp
37	LHBP	577	517	514	0.893	89.08%	93.59%	-4.51 pp
38	EGGW	552	527	523	0.953	94.75%	95.83%	-1.09 pp
39	EDDH	532	509	506	0.968	95.11%	97.74%	-2.63 pp
40	GCLP	521	0	0	0.000	0.00%	0.00%	+0.00 pp
41	LLBG	516	432	430	0.835	83.33%	88.37%	-5.04 pp
42	LPPR	512	480	479	0.936	93.55%	93.75%	-0.20 pp
43	LEIB	509	498	498	0.980	97.84%	96.27%	+1.57 pp
44	EDDK	509	474	474	0.937	93.12%	95.09%	-1.96 pp
45	LROP	509	486	485	0.955	95.28%	97.25%	-1.96 pp
46	EGBB	466	419	418	0.897	89.70%	92.70%	-3.00 pp
47	LIML	465	457	457	0.996	98.28%	97.63%	+0.65 pp
48	LIME	460	447	444	0.965	96.52%	98.04%	-1.52 pp
49	LIRN	447	1	0	0.000	0.00%	0.00%	+0.00 pp
50	LFML	445	440	439	0.996	98.65%	99.10%	-0.45 pp
51	EDDS	433	409	408	0.954	94.23%	95.61%	-1.39 pp
52	LGIR	430	0	0	0.000	0.00%	0.00%	+0.00 pp
53	LFLL	412	398	397	0.998	96.36%	97.33%	-0.97 pp
54	LIPZ	409	386	384	0.941	93.89%	95.35%	-1.47 pp
55	LPFR	407	403	402	0.993	98.77%	98.77%	+0.00 pp
56	LYBE	395	364	364	0.922	92.15%	94.68%	-2.53 pp
57	LICC	391	335	307	0.785	78.52%	4.86%	+73.66 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
58	LIPE	376	371	371	0.989	98.67%	98.94%	-0.27 pp
59	LTAC	374	1	0	0.000	0.00%	0.00%	+0.00 pp
60	LCLK	367	350	347	0.951	94.55%	96.73%	-2.18 pp
61	EGGD	360	334	334	0.933	92.78%	95.28%	-2.50 pp
62	LEVC	357	348	348	0.978	97.48%	98.32%	-0.84 pp
63	GMMN	357	253	253	0.717	70.87%	74.51%	-3.64 pp
64	ENBR	357	341	340	1.006	95.24%	91.32%	+3.92 pp
65	EPKK	356	332	330	0.927	92.70%	85.96%	+6.74 pp
66	LFSB	352	340	340	0.969	96.59%	97.73%	-1.14 pp
67	LGRP	346	315	314	0.910	90.75%	91.04%	-0.29 pp
68	EGPF	346	301	300	0.876	86.71%	88.15%	-1.45 pp
69	LTBJ	341	8	8	0.023	2.35%	0.00%	+2.35 pp
70	LMML	326	318	317	0.975	97.24%	97.85%	-0.61 pp
71	LATI	323	0	0	0.000	0.00%	0.00%	+0.00 pp
72	GCTS	323	0	0	0.000	0.00%	0.00%	+0.00 pp
73	GCXO	320	31	31	0.100	9.69%	0.00%	+9.69 pp
74	LICJ	314	133	133	0.424	42.36%	37.26%	+5.10 pp
75	EBCI	303	296	296	0.977	97.69%	98.68%	-0.99 pp
76	BIKF	296	90	90	0.307	30.41%	33.11%	-2.70 pp
77	ELLX	294	271	271	0.925	92.18%	96.60%	-4.42 pp
78	LFBO	290	288	288	1.000	99.31%	98.97%	+0.34 pp
79	LGTS	281	255	255	0.907	90.75%	77.94%	+12.81 pp
80	LTBS	280	1	1	0.004	0.36%	0.00%	+0.36 pp
81	GMMX	279	13	12	0.043	4.30%	0.00%	+4.30 pp
82	EVRA	278	226	226	0.813	81.29%	81.65%	-0.36 pp
83	LEZL	276	264	264	0.960	95.65%	99.28%	-3.62 pp
84	GCRR	269	0	0	0.000	0.00%	0.00%	+0.00 pp
85	LBSF	267	8	7	0.026	2.62%	0.00%	+2.62 pp
86	EDDP	261	235	235	0.904	90.04%	93.49%	-3.45 pp
87	HEGN	258	0	0	0.000	0.00%	0.00%	+0.00 pp
88	LFRS	257	251	246	0.957	95.72%	96.50%	-0.78 pp
89	UGTB	257	46	45	0.179	17.51%	10.12%	+7.39 pp

#	Aerodrome	Movements (truth)	Found	Correct	Count ratio	Recall	Recall, legacy	Δ recall
90	EPKT	257	240	239	0.930	93.00%	91.83%	+1.17 pp
91	EGNX	255	231	231	0.906	90.59%	93.73%	-3.14 pp
92	LIEO	252	74	73	0.290	28.97%	9.92%	+19.05 pp
93	LGKR	247	12	11	0.045	4.45%	0.00%	+4.45 pp
94	EGAA	246	234	190	0.772	77.24%	86.99%	-9.76 pp
95	EPGD	244	5	1	0.004	0.41%	0.00%	+0.41 pp
96	ENZV	236	67	65	0.284	27.54%	25.85%	+1.69 pp
97	LEBB	235	0	0	0.000	0.00%	0.00%	+0.00 pp
98	LTFE	232	198	198	0.875	85.34%	75.00%	+10.34 pp
99	HECA	231	62	51	0.221	22.08%	1.73%	+20.35 pp
100	LIBD	229	1	0	0.000	0.00%	0.00%	+0.00 pp

22.3 Across the hundred

Role	Movements in truth	Recall, shipped	Recall, legacy	Aerodromes improved	Median count ratio
Departures	65,382	91.52%	82.05%	56 of 100	0.987
Arrivals	65,360	79.48%	79.11%	26 of 100	0.904

A count ratio below one means the aerodrome is under-counted — the usual case, since every abstention is a movement not attributed. Above one means movements are being attributed to an aerodrome that did not see them, which is the more damaging error and the one the scoring penalises at rate k .

The departure row is the result: recall rises by nearly nine points and the median aerodrome is within about one percent of its true movement count.

The arrival row must be read with Section 14 in hand. Recall reads 79.48% against legacy's 79.11%, improving on 26 of 100 — which looks like a mild regression and is not one. Measured as an algorithm on a trend-only flight list, arrivals gain; the shortfall here comes from the merged list also containing departure-only tracks, which this table pairs with reference flights by start time, so some arrivals are counted against a track that carries none. The measurement moved; the data did not. It is left in the table rather than corrected away, because a benchmark that quietly adjusts its own alignment to flatter a result is worth less than one that shows where it is fragile.